

Well Probe V2

Mechanical float switch for dry run protection of LORENTZ solar pumps

The well probe provides a reliable method of run dry protection for LORENTZ pumps. The well probe detects that water is present within a well, tank or other water source. The well probe is typically attached to the riser pipe above the pump and connected to the controller. When the well probe becomes dry (water level is below the probe) the pump switches off to avoid dry running.

Order Information

Item no.: 19-000005 Product name: Well probe sensor V2

Features

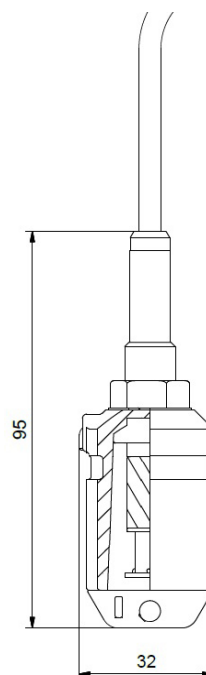
- Reliable dry run protection
- Simple to install using 3 cable ties
- Improved tolerance to dirt
- Splicing kit and cable ties for fixing are included

Technical Data

- Max. operating temperature 55°C
- Enclosure class: IP68
- Submersion depth: max 50 m (164 ft)
- Cable length: 1.5 m
- Wire size: 2 x 0.50 mm² or AWG 20, waterproofed
- Must be mounted in a vertical position
- Meets the requirements for CE

Dimensions / Weight

- Packaging dimensions: 255 x 170 x 40 mm
10.0 x 6.7 x 1.6 in
- Total weight: 0.1 kg / 0.2 lbs



Water Sensor

Device for Dry Run Protection in Applications with LORENTZ Solar Surface Pump Systems

The device can be used to detect the water level in the suction side of the pipeline. When the two electrodes run dry, the LORENTZ Controller will stop the pump and indicates Source Low LED.

ORDER INFORMATION

- **Item no.:** 19-000001 **product name:** Water sensor, pipe inst.
- **Item no.:** 19-000002 **product name:** SS adapter
- **Item no.:** 19-000003 **product name:** Steel adapter

Pipeline adapter must be ordered separately
(available as steel or stainless steel version)

FEATURES

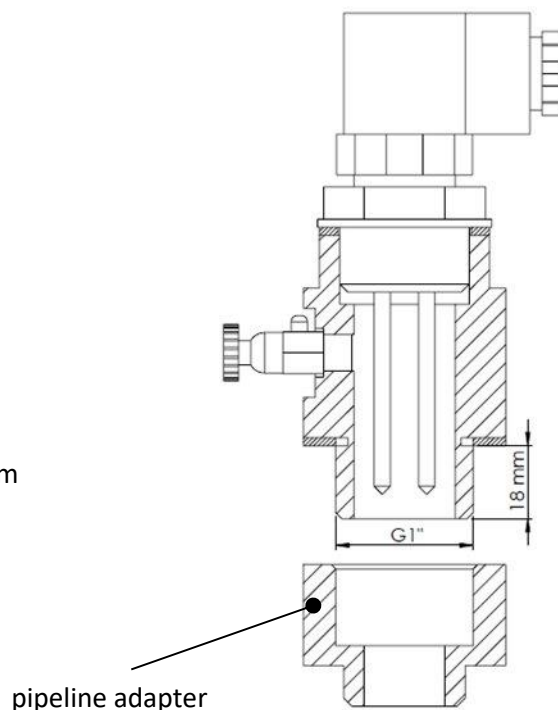
- Reliable dry run protection
- Trouble free operation

TECHNICAL DATA

- Max. operating temperature 70 °C
- Enclosure class: IP54
- Cable length: 4 m
- Wire size: 2x 1.0mm² or AWG 17, waterproofed
- Mounted in vertical position
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packaging dimensions: 180 x 180 x 80 mm
7.1 x 7.1 x 3.1in
- Total weight: 3.5kg / 7.7lbs



Float Switch

Mechanically Activated Device for Water Level Detection in Applications with LORENTZ Solar Pump Systems

The switch can be used to detect the water level within a tank. When the water level in the tank reaches the maximum, the LORENTZ Controller will stop the pump and indicates Tank Full LED.

ORDER INFORMATION

- Item no.: 19-000030 product name: Float Switch

FEATURES

- **N.O.** (normally open) and **N.C.** (normally closed) function
- Reliable water level detection
- Simple to install
- Trouble free operation
- Not sensitive to rotation
- Corrosion-free

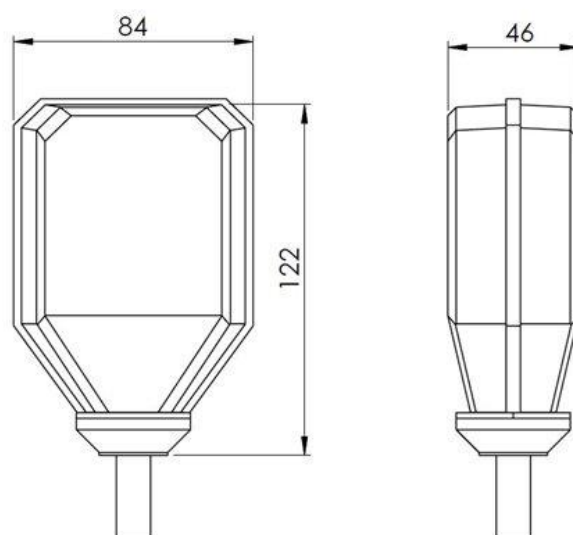


TECHNICAL DATA

- Operating temperature: -10°C to 55°C
- Storage temperature: : -10°C to 55°C
- Enclosure class: IP68
- Cable length: 3m, waterproof
- Wire size: 3x 1.0mm² or AWG 18
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packaging dimensions: 230 x 160 x 55 mm
9.1 x 6.3 x 2.2 in
- Total weight: 0.8 kg / 1.8 lbs



Pressure Switch

Device for Water Pressure Detection in Applications with LORENTZ Solar Pump Systems

The switch can be used to detect the water level within a tank. When the water level in the tank reaches the maximum, the pressure will increase and the LORENTZ Controller will stop the pump and indicates Tank Full LED.

ORDER INFORMATION

- **Item no.:** 19-000310 **product name:** Pressure Switch 1-5bar

FEATURES

- Reliable water pressure detection
- Simple to install
- Trouble free operation
- Corrosion-free

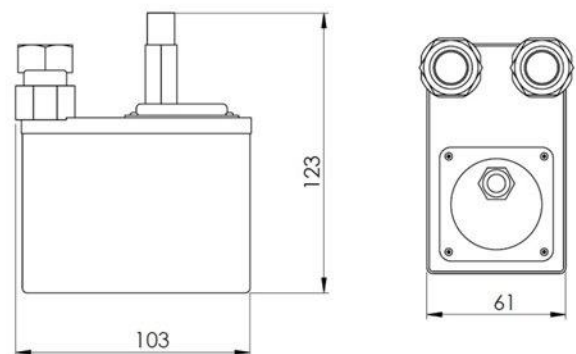
TECHNICAL DATA

- Pressure range between 1 and 5 bar
- 2x cable gland Ø 5-9mm
- G 1/4" pipe thread
- Meets the requirements for CE



DIMENSION/WEIGHT

- Packaging dimensions: 130 x 105 x 65 mm
5.1 x 4.2 x 2.6 in
- Total weight: 0.5 kg / 1 lbs



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Liquid Level Sensor

Sensor for measuring the level of liquid in a well or tank

USE / PURPOSE

The range of liquid level sensors use pressure to measure the liquid level in a well or tank. The sensors can be used for collection of long term liquid level monitoring and also for pump control in applications where a well probe cannot be used. The liquid level sensor must be used with a compatible LORENTZ Controller (see requirements).

FEATURES

- Gauge pressure sensor, pressure measurement relative to atmosphere
- For measuring levels of liquid in a well or tank
- Accurate, robust sensor
- For use with LORENTZ PumpScanner and pumpMANAGER

REQUIREMENTS

- LORENTZ PS2 controller, LORENTZ PSk2 controller or PS Controller equipped with a licensed PS DataModule
- The cable contains a capillary tube to balance the sensor against atmospheric changes. The sensor must be ordered with the required cable length for the application as it cannot be spliced. Further cable lengths are available on demand, please contact LORENTZ.

TECHNICAL DATA

- Sensor type: 2 wire gauge sensor
- Enclosure class: IP68
- Sensor housing: stainless steel
- Connects to LORENTZ PS DataModule
- Oil resistant cable with capillary tube
- Overpressure: 2x full scale
- Output signal :4-20mA output
- Voltage : 12-28VDC
- Application temp.: -10 to 80°C
14 to 176°F
- Accuracy class :0.5% FS
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packing dimensions:
19-005040: 350 x 350 x 100 mm / 13.8 x 13.8 x 3.9 in
19-005050: 350 x 350 x 100 mm / 13.8 x 13.8 x 3.9 in
19-005080: 350 x 350 x 100 mm / 13.8 x 13.8 x 3.9 in
19-004395: 350 x 350 x 180 mm / 13.8 x 13.8 x 7.1 in
- Weight:
19-005040: 4 kg / 8.8 lbs
19-005050: 5 kg / 11.0 lbs
19-005080: 7 kg / 15.4 lbs
19-004395: 11 kg / 24.3 lbs



ORDER INFORMATION

Item #	Product	Pressure range	Cable length
19-005040	Liquid Level Sensor, 0-10m/33ft, 30m/100ft Cable	0-100 kPa 0 to 10 m / 0 to 33 ft 0 to 1 bar / 0 to 14.5 psi	30 m / 100 ft *
19-005050	Liquid Level Sensor, 0-20m/66ft, 40m/130ft Cable	0-200 kPa 0 to 20 m / 0 to 66 ft 0 to 2 bar / 0 to 29 psi	40 m / 130 ft *
19-005080	Liquid Level Sensor, 0-50m/164ft, 60m/200ft Cable	0-500 kPa 0 to 50 m / 0 to 164 ft 0 to 5 bar / 0 to 72.5 psi	60 m / 200 ft *
19-004395	Liquid Level Sensor, 0-100m/328ft, 140m/460ft Cable	0-1000 kPa 0 to 100 m / 0 to 328 ft 0 to 10 bar / 0 to 145 psi	140 m / 460 ft *

*Further cable length is available on demand. Please contact LORENTZ.

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Liquid Pressure Sensor

Sensor for measuring the pressure of liquid in a pipe or vessel

USE / PURPOSE

The sensors are commonly used to measure the pressure in the delivery pipeline or in a vessel. The pressure signal is used with a suitable LORENTZ pump controller to measure pressure and for pressure switching or constant pressure applications. The liquid pressure sensor must be used with a compatible LORENTZ Controller (see requirements).

FEATURES

- Gauge pressure sensor, pressure measurement relative to atmosphere
- For measuring pressure in a pipe or vessel
- Accurate, robust sensor
- For use with LORENTZ PumpScanner and pumpMANAGER

REQUIREMENTS

- LORENTZ PS2 controller, LORENTZ PSk2 controller or PS Controller equipped with a licensed PS DataModule
- Care must be taken to position the sensor without turbulent water to ensure accurate measurement
- G1/4" or G1/2" female threaded filling / air vent hole is required to mount the sensor



TECHNICAL DATA

- Sensor type: 2 wire gauge sensor
- Enclosure class: IP65
- Sensor housing : stainless steel
- Connects to LORENTZ PS DataModule
- 5m/10m (16ft/33ft) cable length
- Overpressure: 1.5x full scale
- Output signal : 4-20 mA
- Voltage : 11-28 VDC
- Application temp.: -30 to 80°C
-22 to 176°F
- Accuracy class : 0.5% full scale
- Thread type: G1/2" male (G1/4" with adapter)
- Meets the requirements for CS

ORDER INFORMATION

Item #	Product	Pressure range
19-004450	Liquid Pressure Sensor, LPS-500	0-500 kPa 0 to 5 bar / 0 to 72.5 psi
19-004460	Liquid Pressure Sensor, LPS-1000	0-1000 kPa 0 to 10 bar / 0 to 145 psi
19-002760	Liquid Pressure Sensor, LPS-2000	0-2000 kPa 0 to 20 bar / 0 to 290 psi

DIMENSION/WEIGHT

- Packing dimensions: 190 x 180 x 80 mm
7.5 x 7.1 x 3.2 in
- Weight: 0.6 kg / 1.3 lbs

Water Meter

Device to measure the water flow rate

ORDER INFORMATION

- **Item no.:** 19-002160 **product name:** Water Meter, WP-DN40, 0.1cbm/p
- **Item no.:** 19-002165 **product name:** Water Meter, WP-DN50, 0.1cbm/p
- **Item no.:** 19-002170 **product name:** Water Meter, WP-DN65, 0.1cbm/p
- **Item no.:** 19-002180 **product name:** Water Meter, WP-DN80, 0.1cbm/p
- **Item no.:** 19-002190 **product name:** Water Meter, WP-DN100, 0.1cbm/p
- **Item no.:** 19-002200 **product name:** Water Meter, WP-DN125, 0.1cbm/p
- **Item no.:** 19-002210 **product name:** Water Meter, WP-DN150, 1cbm/p
- **Item no.:** 19-002202 **product name:** Water Meter, WP-DN200, 1cbm/p

FEATURES

- Dry dial register ensures clear reading
- Low pressure loss, long working life
- Easy to install
- **Only for use in combination with DataModule:**
 - The water meter is equipped with a reed switch assembly which can be connected to the DataModule



TECHNICAL DATA

- Water temperature: max. 40°C
- Water pressure: max. 16 bar
- IP64
- Meets the requirements for CE

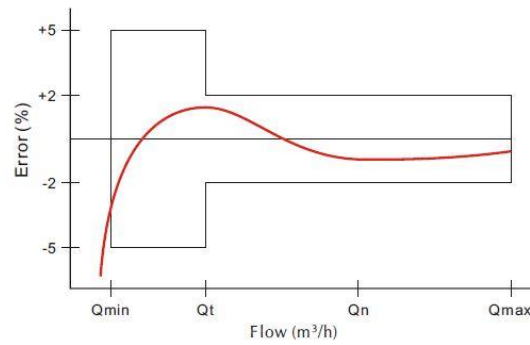
Nominal Diameter	DN	40	50	65	80	100	125	150	200
Max. flow rate [m³/h]	Q _{max}	20	30	50	80	120	200	300	500
Nominal flow rate [m³/h]	Q _n	10	15	25	40	60	100	150	250
Transition flow rate [m³/h]	Q _t	3.0	3.0	5.0	8.0	12	20	30	50
Minimum flow rate [m³/h]	Q _{min}	0.7	0.7	0.75	1.2	1.8	3.0	4.5	7.5

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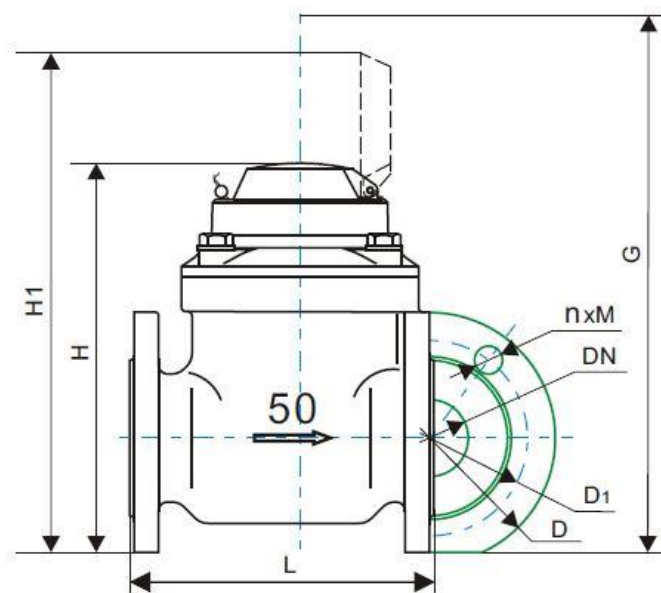
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DIMENSION/WEIGHT

Nominal Diameter	DN	40	50	65	80	100	125	150	200
Length [mm]	L	260	200	200	225	250	250	300	350
Height [mm]	H	225	232	242	252	262	275	325	355
Working height [mm]	H1	303	303	313	323	333	346	396	426
Height [mm]	G	360	360	360	360	360	360	420	420
Outside diameter [mm]	D	150	165	185	200	220	250	285	340
Circle diameter [mm]	D1	110	125	145	160	180	210	240	295
Connecting bolt quantity		4xM16	4xM16	4xM16	8xM16	8xM16	8xM16	8xM20	12xM20
Weight [kg]		12	12	13	16	18	20	42	74



SunSwitch

Automatic Solar Level Switching Device

The SunSwitch can be used for any application where you need to switch depending on the solar intensity. It is an autonomous working unit that supplies itself with power from the PV-module on its rear side.

ORDER INFORMATION

- Item no.: 19-000050 **product name:** SUN SWITCH Light-Sensor

FEATURES

- Trouble free operation
- Pump will only run when minimum flow rates or pressure levels can be achieved which are required by certain applications
- Reduced wear on the pump and motor since the pump only starts when it will continue to run

TECHNICAL DATA

- Contact rating:
 - Max. voltage 250 VAC / 220 VDC
 - Max. current 2A
 - Max. power 60 W / 62.5 VA
- Enclosure class: IP54
- Cable length: 4 m
- Wire size: 3x 1.0mm² or AWG 18, waterproofed
- Mounted close to the solar array
- Manual with further information is available in PartnerNet
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packaging dimensions: 230 x 115 x 90 mm
9.1 x 4.5 x 3.5 in
- Total weight: 1.2 kg / 0.6 lbs



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Surge Protector

Device to Protect LORENTZ Pump Accessories from Voltage Spikes

ORDER INFORMATION

- Item no.: 19-000280 product name: Surge Protector

FEATURES

- Reliable surge protection for all LORENTZ pump accessories
- Can be installed inside the PS Controller

TECHNICAL DATA

- Max. voltage: 14 VDC
- Max current 8/20 μ s: 500 A
- Enclosure class: IP65
- Ambient temperature: max. 50°C
- Wire size: 2x 1.5mm² or AWG 16
- Meets the requirements for CE



DIMENSION/WEIGHT

- Packing dimensions: 70 x 45 x 20 mm
 2.8 x 1.8 x 0.8 in
- Total weight 0.1 kg / 0.2 lbs

Cable Splice Kit 2.5 to 6 mm²

Kit for Permanently Insulate and Seal Cable Splice

ORDER INFORMATION

- **Item no.:** 19-000010 **product name:** Cable splice kit 2.5-6sqmm

FEATURES

- Efficient watertight seal
- Simple to install
- High quality components
- Kit for wire sizes from 2.5 to 6 mm²

INCLUDED IN DELIVERY

Four splices per kit:

- 4 pcs. crimp sleeves
- 4 pcs. small shrink sleeves
- 1 pc. large shrink sleeve
- Assembly instruction

DIMENSION/WEIGHT

- Packaging dimensions: 400 x 120 x 30 mm
15.8 x 4.7 x 1.2 in
- Total weight: 0.1 kg / 0.2 lbs



Cable Splice Kit 6 to 10 mm²

Kit for Permanently Insulate and Seal Cable Splice

ORDER INFORMATION

- **Item no.:** 19-000020 **product name:** Cable splice kit 6-10sqmm

FEATURES

- Efficient watertight seal
- Simple to install
- High quality components
- Kit for wire sizes from 6 to 10 mm²

INCLUDED IN DELIVERY

Four splices per kit:

- 4 pcs. crimp sleeves
- 4 pcs. small shrink sleeves
- 1 pc. large shrink sleeve
- Assembly instruction

DIMENSION/WEIGHT

- Packaging dimensions: 400 x 120 x 30 mm
15.8 x 4.7 x 1.2 in
- Total weight: 0.1 kg / 0.2 lbs



PV Disconnect 440-40-1

Connection box with DC disconnect

Description

An outdoor rated, combining connection box with DC disconnect switch that allows 1 strings of PV modules to be connected safely to a solar pump system.

The PV disconnect is also designed to accept an optional lightning protection device.

Features

- DC rated disconnect to provide safe isolation of the system
- Robust weather proof housing designed to make installation simple
- Lockable to secure the system during maintenance (power locked off)
- For professional installation of pumping systems
- Internal touch protection with screws
- Designed to be used with LORENTZ PS2-150 to PS2 4000 systems



photo may differ from actual product

Ordering and shipping information

- Item no: 19-000125
- Product name: PV Disconnect 440-40-1
- Packed volume 0.01 m³ (0.35 ft³)
- Packed weight 1.9 kg (4.2 lbs)

Approvals and standards

- Switch IEC 60947-3



Technical data / Specifications

Maximum voltage	440 V DC	
Maximum current per string	40 A	
Maximum total current	40 A	
Number of strings.	1	
Input cables	4 - 10 mm ²	AWG 12 - 8
Output cables	4 - 10 mm ²	AWG 12 - 8
PG glands (input)	2 x M16	
PG glands (output)	2 x M16	
Lightening protection mounting hole	PG16 cap	
Environmental protection	IP68	NEMA6
Housing material	Polycarbonate	

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Optional lightning surge protector

Provides protection for the pump controller from incoming high voltages on the PV side. The surge protector connects through a pre-drilled and blanked mounting hole in the PV connect housing

- Proper grounding of the device is required to achieve protection
- Item no.: 19-002120 MNSPD-115 PS2-150 to PS2-200
- Item no.: 19-002130 MNSPD-300 PS2-600 to PS2-1800
- Item no.: 19-002140 MNSPD-600 PS2-4000



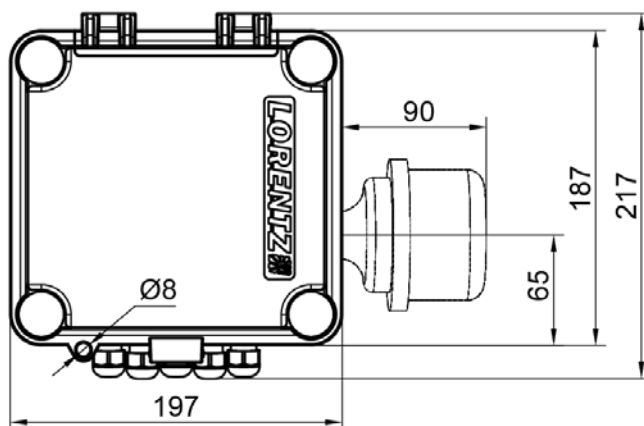
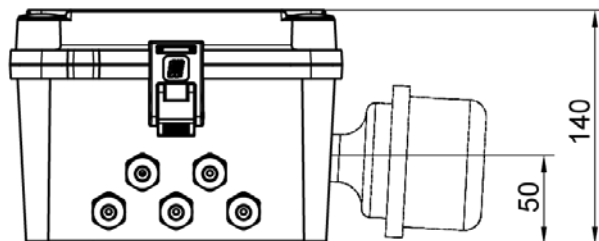
Mounting options

- Wall mount using 4 holes with weather protection
- Designed for optional pole mounting. Mounting points are pre-marked inside the housing.



Dimensions and weight

- See diagram for mm sizes
- Max height 220 mm (8.66")
- Max width (no surge protector) 197 mm (7.75")
- Max width (surge protector) 297 mm (Max 11.7")
- depth 140 mm (5.5")
- Weight 1403 g (3.1 lbs)



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PV Disconnect 440-40-3

Connection box with DC disconnect

Description

An outdoor rated, combining connection box with DC disconnect switch that allows up to 3 strings of PV modules to be connected safely to a solar pump system.

The PV disconnect is also designed to accept an optional lightning protection device.

Features

- DC rated disconnect to provide safe isolation of the system
- Combining function, allows 3 strings to be connected to the solar pump system
- Robust weather proof housing designed to make installation simple
- Lockable to secure the system during maintenance (power locked off)
- For professional installation of pumping systems
- Internal touch protection with screws
- Designed to be used with LORENTZ PS2-150 to PS2 4000 systems

Ordering and shipping information

- Item no: 19-000126
- Product name: PV Disconnect 440-40-3
- Packed volume: 0.01 m³ (0.35 ft³)
- Packed weight: 2.1 kg (4.6 lbs)

Approvals and standards

- Switch IEC 60947-3



photo may differ from actual product

Technical data / Specifications

Maximum voltage	440 V DC	
Maximum current per string	15 A	
Maximum total current	40 A	
Number of strings.	3	
Input cables	2.5 - 4 mm ²	AWG 14 - 12
Output cables	4 - 10 mm ²	AWG 12 - 8
PG glands (input)	6 x M16	
PG glands (output)	2 x PG11	
Lightening protection mounting hole	PG16 cap	
Environmental protection	IP68	NEMA6
Housing material	Polycarbonate	

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Optional lightning surge protector

Provides protection for the pump controller from incoming high voltages on the PV side. The surge protector connects through a pre-drilled and blanked mounting hole in the PV connect housing

- Proper grounding of the device is required to achieve protection
- Item no.: 19-002120 MNSPD-115 PS2-150 to PS2-200
- Item no.: 19-002130 MNSPD-300 PS2-600 to PS2-1800
- Item no.: 19-002140 MNSPD-600 PS2-4000



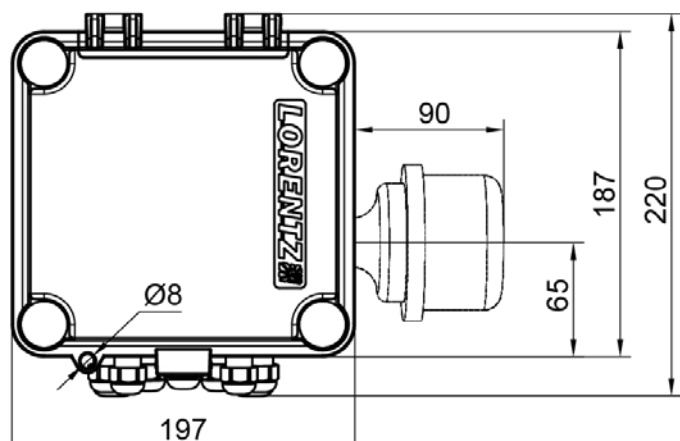
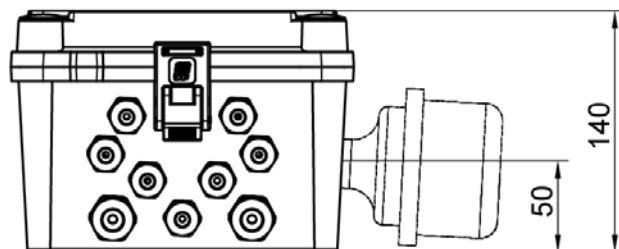
Mounting options

- Wall mount using 4 holes with weather protection
- Designed for optional pole mounting. Mounting points are pre-marked inside the housing.



Dimensions and weight

- See diagram for mm sizes
- Max height 220 mm (8.66")
- Max width (no surge protector) 197 mm (7.75")
- Max width (surge protector) 297 mm (Max 11.7")
- depth 140 mm (5.5")
- Weight 1.62 kg (3.56 lbs)



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PV Disconnect 440-40-6

Connection box with DC disconnect

Description

An outdoor rated, combining connection box with DC disconnect switch that allows up to 6 strings of PV modules to be connected safely to a solar pump system.

The PV disconnect is also designed to accept an optional lightning protection device.

Features

- DC rated disconnect to provide safe isolation of the system
- Combining function, allows 6 strings to be connected to the solar pump system
- Robust weather proof housing designed to make installation simple
- Lockable to secure the system during maintenance (power locked off)
- For professional installation of pumping systems
- Internal touch protection with screws
- Designed to be used with LORENTZ PS2-150 to PS2 4000 systems



photo may differ from actual product

Ordering and shipping information

- Item no: 19-000124
- Product name: PV Disconnect 440-40-6
- Packed volume: 0.015 m³ (0.54 ft³)
- Packed weight: 2.7 kg (6 lbs)

Approvals and standards

- Switch IEC 60947-3



Technical data / Specifications

Maximum voltage	440 V DC	
Maximum current per string	10 A	
Maximum total current	40 A	
Number of strings.	6	
Input cables	2.5 - 4 mm ²	AWG 14 - 12
Output cables	4 - 10 mm ²	AWG 12 - 8
PG glands (input)	12 x M16	
PG glands (output)	2 x PG11	
Lightening protection mounting hole	PG16 cap	
Environmental protection	IP68	NEMA6
Housing material	Polycarbonate	

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Optional lightning surge protector

Provides protection for the pump controller from incoming high voltages on the PV side. The surge protector connects through a pre-drilled and blanked mounting hole in the PV connect housing

- Proper grounding of the device is required to achieve protection
- Item no.: 19-002120 MNSPD-115 PS2-150 to PS2-200
- Item no.: 19-002130 MNSPD-300 PS2-600 to PS2-1800
- Item no.: 19-002140 MNSPD-600 PS2-4000



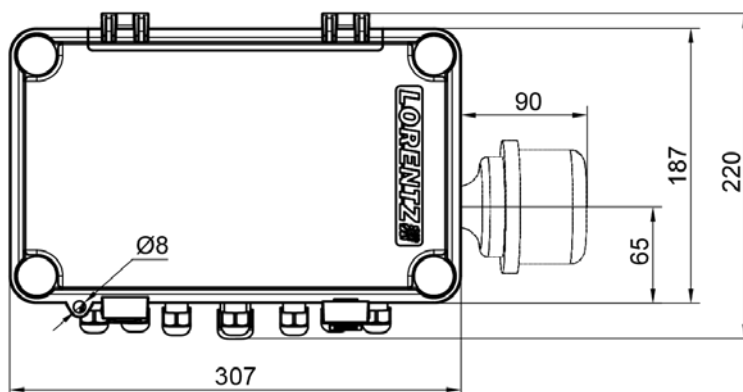
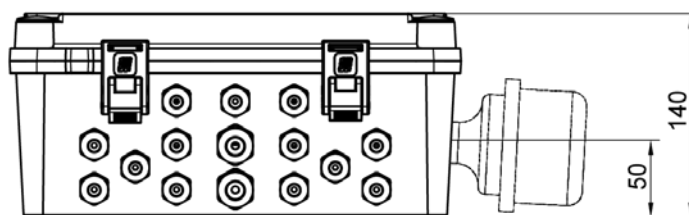
Mounting options

- Wall mount using 4 holes with weather protection
- Designed for optional pole mounting. Mounting points are pre-marked inside the housing.



Dimensions and weight

- See diagram for mm sizes
- Max height 220 mm (8.66")
- Max width (no surge protector) 307 mm (12.1")
- Max width (surge protector) 397 mm (Max 15.6")
- depth 140 mm (5.5")
- Weight 2.13 kg (4.7 lbs)



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PV Disconnect 1000-40-5

Connection box with DC disconnect

Description

An outdoor rated, combining connection box with DC disconnect switch that allows up to 5 strings of PV modules to be connected safely to a solar pump system.

Features

- DC rated disconnect to provide safe isolation of the system
- Combining function, allows 5 strings to be connected to the solar pump system
- Robust weather proof housing designed to make installation simple
- Lockable to secure the system during maintenance (power locked off)
- For professional installation of pumping systems
- Internal touch protection with screws
- Designed to be used with the LORENTZ PSK range of systems



photo may differ from actual product

Ordering and shipping information

- Item no: 19-000121
- Product name: PV Disconnect 1000-40-5
- Packed volume: 0.015 m³ (0.54 ft³)
- Packed weight: 3.2 kg (7.1 lbs)

Approvals and standards

- Switch IEC 60947-3



Technical data / Specifications

Maximum voltage	1000 V DC	
Maximum current per string	40 A	
Maximum total current	40 A	
Number of strings.	5	
Input cables	4 - 10 mm ²	AWG 12 - 8
Output cables	4 - 16 mm ²	AWG 12 - 6
PG glands (input)	10 x M16	
PG glands (output)	2 x PG11	
Environmental protection	IP68	NEMA6
Housing material	Polycarbonate	

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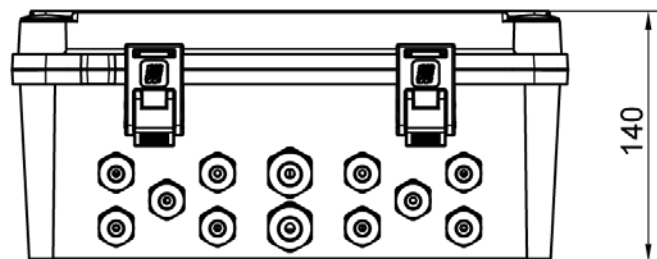
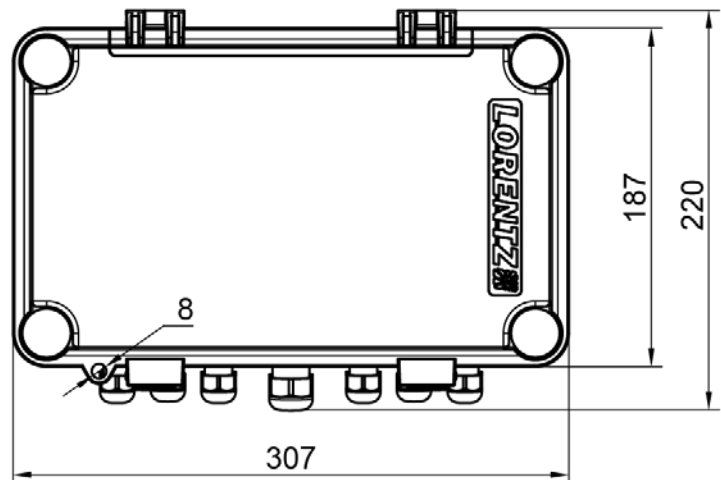
Mounting options

- Wall mount using 4 holes with weather protection
- Designed for optional pole mounting. Mounting points are pre-marked inside the housing.



Dimensions and weight

- See diagram for mm sizes
- Max height 220 mm (8.66")
- Max width 307 mm (12.1")
- depth 140 mm (5.5")
- Weight 2.67 kg (5.9 lbs)



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PV Combiner 1000-125-4

Connection box or parallel PV wiring

Description

An outdoor rated, combining connection box that allows up to 4 strings of PV modules to be connected safely to a solar pump system.

This product is typically used to combine strings from multiple LORENTZ 1000-40-5 PV Disconnects.

Features

- Combining function, allows 4 strings to be connected to the solar pump system
- Robust weather proof housing designed to make installation simple
- Lockable for additional protection
- Tool required to open the housing
- Designed to be used with LORENTZ PSk range of systems



photo may differ from actual product

Ordering and shipping information

- Item no: 19-000122
- Product name: PV Combiner 1000-125-4
- Packed volume: 0.015 m³ (0.54 ft³)
- Packed weight: 2.75 kg (6.1 lbs)

Approvals and standards



Technical data / Specifications

Maximum voltage	1000 V DC	
Maximum current per string	125 A	
Maximum total current	125 A	
Number of strings.	4	
Input cables	10 - 35 mm ²	AWG 8 - 2
Output cables	10 - 35 mm ²	AWG 8 - 2
PG glands (input)	8 x PG16	
PG glands (output)	2 x PG16	
Environmental protection	IP68	NEMA6
Housing material	Polycarbonate	

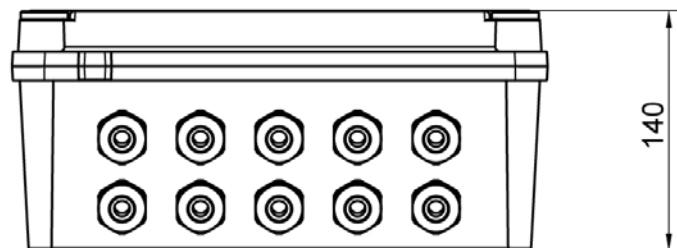
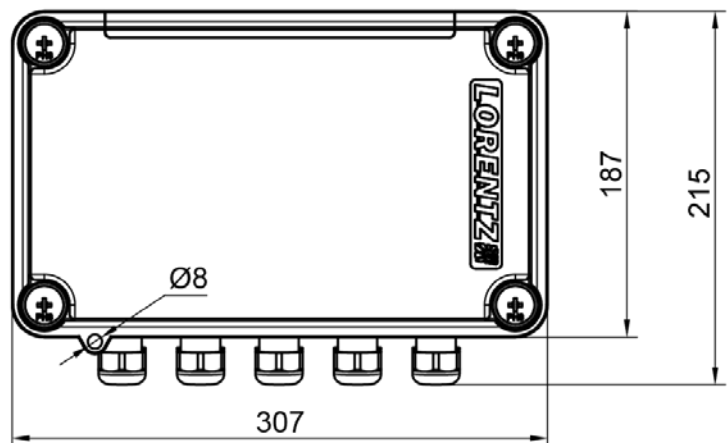
Mounting options

- Wall mount using 4 holes with weather protection
- Designed for optional pole mounting. Mounting points are pre-marked inside the housing.



Dimensions and weight

- See diagram for mm sizes
- Max height 215 mm (8.66")
- Max width 307 mm (12.1")
- depth 140 mm (5.5")
- Weight 2.2 kg (4.8 lbs)



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PV Protect 1000-125

Surge protective device for PV systems

Description

An outdoor rated, surge protection device for PSk solar pump systems. The PV Protect 1000-125 provides an enhanced level of protection from electrical surges from the PV generator typically caused by indirect lightning.

This product is installed between the PV generator and the pump controller.

Features

- Provides enhanced protection to the pump controller from incoming voltage surges
- Robust weather proof housing designed to make installation simple
- Tool required to open the housing
- Lockable for additional protection
- Correct grounding is required to make this device effective
- Designed to be used with LORENTZ PSk range of systems



photo may differ from actual product

Ordering and shipping information

- Item no: 19-000123
- Product name: PV Protect 1000-125
- Packed volume: 0.015 m³ (0.54 ft³)
- Packed weight: 2.2 kg (4.9 lbs)

Approvals and standards



Technical data / Specifications

Maximum voltage	1000 V DC	
Maximum current per string	125 A	
Maximum total current	125 A	
Input cables	10 - 35 mm ²	AWG 8 - 2
Output cables	10 - 35 mm ²	AWG 8 - 2
Ground cable	16 mm ²	AWG 6
PG glands (input)	2 x PG16	
PG glands (output)	2 x PG16	
Environmental protection	IP68	NEMA6
Housing material	Polycarbonate	

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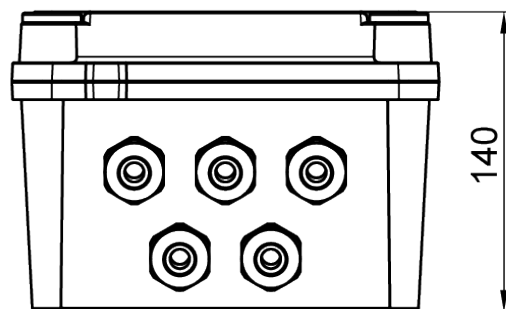
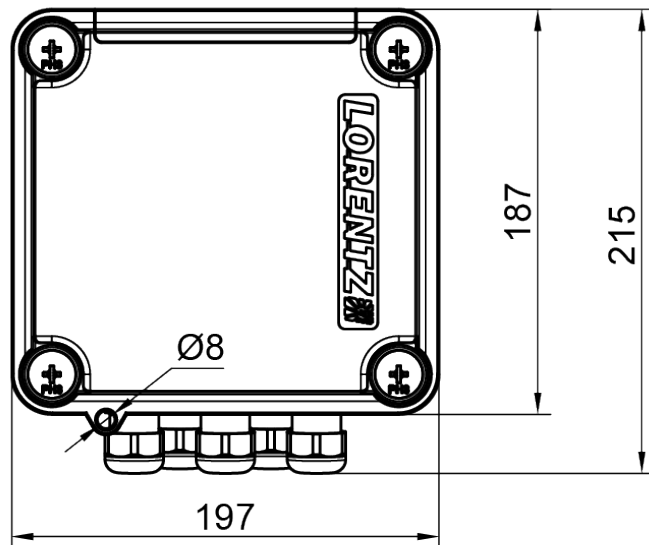
Mounting options

- Wall mount using 4 holes with weather protection
- Designed for optional pole mounting. Mounting points are pre-marked inside the housing.



Dimensions and weight

- See diagram for mm sizes
- Max height 215 mm (8.66")
- Max width 197 mm (7.8")
- depth 140 mm (5.5")
- Weight 1.7 kg (3.7 lbs)



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Sacrificial Anode for HR Pump

Zinc sacrificial anode for corrosion protection of HR pump ends

USE / PURPOSE

The LORENTZ Sacrificial Anode is designed for use with LORENTZ pump ends. Where solar pump ends operate in aggressive water conditions acidic water can attacks the passive layer of the high quality-steel pump end. This corrosion reduces the life time of the product. The LORENTZ Sacrificial Anode protects the pump system from this type of corrosion and extends the pump life

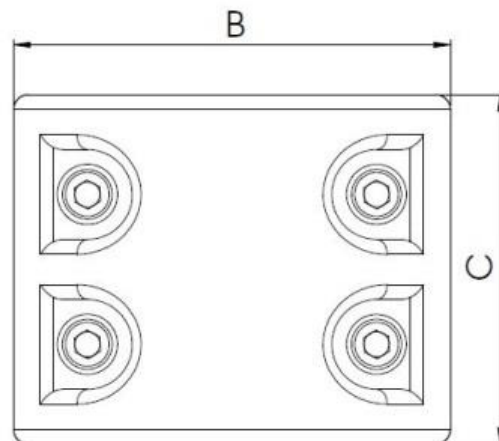
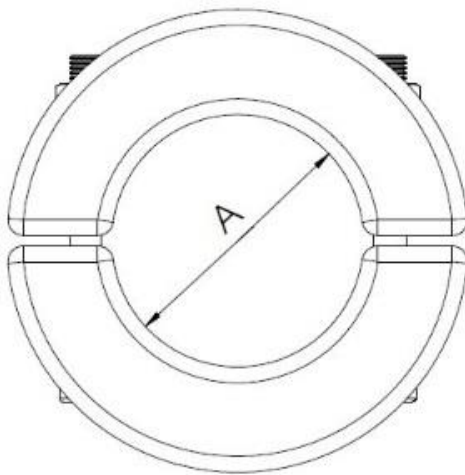
ORDER INFORMATION

- Item no.: 19-002621 Product name: Anode, Sacrificial Zink Anode, Di42mm
- Item no.: 19-002622 Product name: Anode, Sacrificial Zink Anode, Di48mm

FEATURES

- Simple and cost effective part to protect the pump end from corrosion
- Designed for HR pump ends

TECHNICAL DATA



	A	B	C	Weight	Surface	Item no.
HR03/04/05	48 mm 1,89 in	88 mm 3,46 in	70 mm 2,76 in	1,76 kg 3,88 lbs	59100 mm ² 91,61 in ²	19-002622
HR07/10/14/23/32	42 mm 1,65 in	82 mm 3,23 in		1,68 kg 3,70 lbs	55500 mm ² 86,01 in ²	19-002621

Sacrificial Anode for 4" and 6" motors

Zinc sacrificial anode for corrosion protection of Pump Units

Application

The LORENTZ Sacrificial Anode is designed for use with LORENTZ pump units. Where a pump unit operates in aggressive water conditions acidic water can attack the surface of the high quality-stainless steel pump unit. This corrosion reduces the life expectancy of the product. The LORENTZ Sacrificial Anode reduces corrosion at pump end and extends the pump life time.

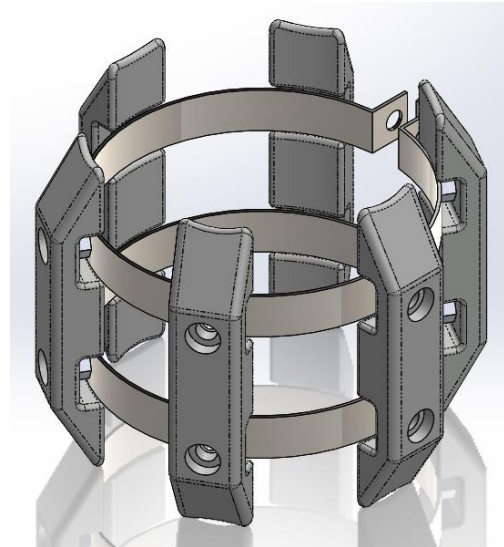
ORDER INFORMATION

- Item no.: 19-002630

Product name: Sacrificial Anode 4" motors

- Item no.: 19-002632

Product name: Sacrificial Anode 6" Motors



picture shows sample of 6" anode

FEATURES

- Simple and cost effective part to reduce corrosion at pump end and motor
- Designed for installation around the body of 4" and 6" motors below the pump end.
- 4" version requires a minimum 6" well / 6" version requires a minimum 8" well
- Zinc anodes
- Stainless steel clamps and screws

TECHNICAL DATA (installed)

Item No	motor	Qty. of anodes	Outer diameter	length	Zinc Weight	Zinc Surface
19-002630	4" motors	4	147 mm 5.78 in	160 mm 6.30 in	1.64 kg 3.63 lbs.	58300 mm ² 90 inch ²
19-002632	6" motors	6	185 mm 7.28 in		2.46 kg 5.43 lbs	87470 mm ² 136 in ²

DIMENSION/WEIGHT

- Packing dimensions: 180 x 180 x 50 mm / 7.1 x 7.1 x 2.0 in
- Total packing weight: 2.0 kg / 4.4 lbs or 2.8kg / 6.2lbs

Sacrificial Anode for 4" and 6" motors

Zinc sacrificial anode for corrosion protection of Pump Units

Application

The LORENTZ Sacrificial Anode is designed for use with LORENTZ pump units. Where a pump unit operates in aggressive water conditions acidic water can attack the surface of the high quality-stainless steel pump unit. This corrosion reduces the life expectancy of the product. The LORENTZ Sacrificial Anode reduces corrosion at pump end and extends the pump life time.

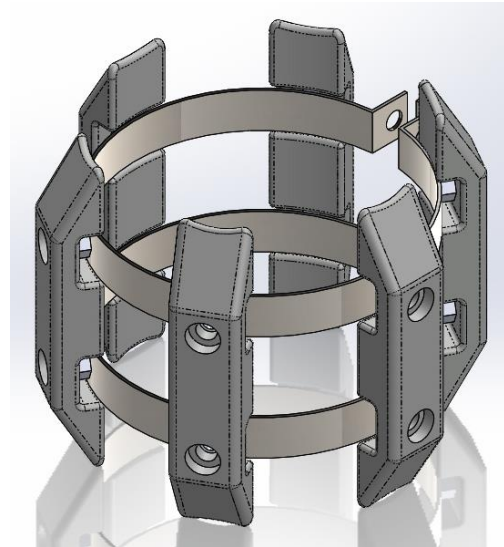
ORDER INFORMATION

- Item no.: 19-002630

Product name: Sacrificial Anode 4" motors

- Item no.: 19-002632

Product name: Sacrificial Anode 6" Motors



picture shows sample of 6" anode

FEATURES

- Simple and cost effective part to reduce corrosion at pump end and motor
- Designed for installation around the body of 4" and 6" motors below the pump end.
- 4" version requires a minimum 6" well / 6" version requires a minimum 8" well
- Zinc anodes
- Stainless steel clamps and screws

TECHNICAL DATA (installed)

Item No	motor	Qty. of anodes	Outer diameter	length	Zinc Weight	Zinc Surface
19-002630	4" motors	4	147 mm 5.78 in	160 mm 6.30 in	1.64 kg 3.63 lbs.	58300 mm ² 90 inch ²
19-002632	6" motors	6	185 mm 7.28 in		2.46 kg 5.43 lbs	87470 mm ² 136 in ²

DIMENSION/WEIGHT

- Packing dimensions: 180 x 180 x 50 mm / 7.1 x 7.1 x 2.0 in
- Total packing weight: 2.0 kg / 4.4 lbs or 2.8kg / 6.2lbs

PP150AC PowerPack

AC/DC Converter to Supply PS2-150 Pump Systems with Backup Power from a Generator or Mains Supply

ORDER INFORMATION

- Item no.: 19-004240 / 19-004260 product name: Power Pack 150

TECHNICAL DATA

Product name	Power Pack 150	
Item no.	19-004260	19-004240
Pump System	PS2-150	
AC input		
Nominal voltage	120 VAC	240 VAC
Max. current	10 A	6 A
Frequency	50-60 Hz	50-60 Hz
DC output		
Max. voltage	24 VDC	24 VDC
Max. current	19.0 A	19.0 A

- AC input: 120/240 V ($\pm 15\%$) as factory setting, 50 to 60 Hz
- DC output: max. 24 V DC (no load), 450 W
- Enclosure: steel, gasket-sealed, indoor-use, hinged front cover with key-lock
- Must be protected from direct mid-day sun
- Enclosure class: IP22
- Ambient temperature: max. 45 °C / 113°F
- PV cable length 2 m, with Y MC4 plugs to connect in parallel to PV-generator
- Main supply AC input cable length 1.8 m, pre-connected grid main supply plugs
- Manual with further information is available on partnerNET
- Meets the requirements for CE



DIMENSION/WEIGHT

- Packaging dimensions: 390 x 270 x 160 mm / 15.4 x 10.6 x 6.3 in
- Total weight: 6.4 kg / 14 lbs

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PP200AC PowerPack

AC/DC Converter to Supply PS2-200 Pump Systems with Backup Power from a Generator or Mains Supply

ORDER INFORMATION

- Item no.: 19-004250 / 19-004270 product name: Power Pack 200

TECHNICAL DATA

Product name	Power Pack 200	
Item no.	19-004270	19-004250
Pump System	PS2-200	
AC input		
Nominal voltage	120 VAC	240 VAC
Max. current	10 A	6 A
Frequency	50-60 Hz	50-60 Hz
DC output		
Max. voltage	48 VDC	48 VDC
Max. current	10.0 A	10.0 A

- AC input: 120/240 V ($\pm 15\%$) as factory setting, 50 to 60 Hz
- DC output: max. 48 V DC (no load), 450 W
- Enclosure: steel, gasket-sealed, indoor-use, hinged front cover with key-lock
- Must be protected from direct mid-day sun
- Enclosure class: IP22
- Ambient temperature: max. 45 °C / 113°F
- PV cable length 2 m, with MC4 plugs
- Main supply AC input cable length 1.8 m, pre-connected grid main supply plugs
- Manual with further information is available on partnerNET
- Meets the requirements for CE



DIMENSION/WEIGHT

- Packaging dimensions: 390 x 270 x 160 mm / 15.4 x 10.6 x 6.3 in
- Total weight: 6.4 kg / 14 lbs

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PP600H AC PowerPack

AC/DC Converter to Supply PS2-600 and specific PS2-1800HR Pump Systems with Backup Power from a Generator or Mains Supply

ORDER INFORMATION

- Item no.: 19-001040 product name: PowerPack 600H, UL

TECHNICAL DATA

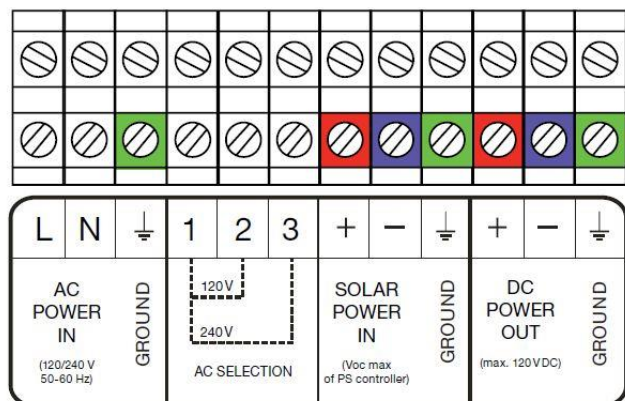
- AC input: 120V or 240V ($\pm 15\%$), selectable, 47 to 63Hz
- AC input current: 6.2A (120V) or 3.1A (240V)
- PV input voltage: V_{mp} : 130V DC, depending on pump controller max. voltages
- PV max open circuit voltage: 200V DC depending on pump controller max. voltages
- DC output: 120V DC (no load), 700W
- Overload protection: internal fuse, 15A
- Enclosure: steel, gasket-sealed, indoor-use, hinged front cover with key-lock
- Must be protected from direct mid-day sun
- Enclosure class: IP22
- Ambient temperature: max. 45°C
- All wired must be #14 AWG (2.5 mm²) or larger
- Manual with further information is available in partnerNET
- Meets requirements for CE

Electrical Safety



DIMENSION/WEIGHT

- 450 x 300 x 140 mm
17.7 x 11.8 x 5.5 in
- Total weight: 16 kg / 35 lbs



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PP2000 AC PowerPack

AC/DC Converter to Supply specific PS2-1800 Pump Systems with Backup Power from a Generator or Mains Supply

ORDER INFORMATION

- Item no.: 19-001050 product name: PowerPack 2000, UL

TECHNICAL DATA

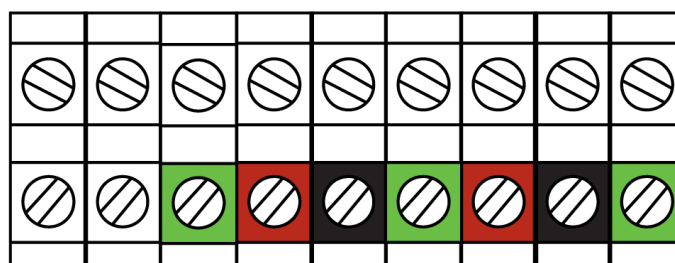
- AC input: 240V ($\pm 15\%$), 47 to 63Hz
- AC input current: 6.2A
- PV input voltage: V_{mp} : 110V -140V DC, depending on pump controller max. voltages
- PV max open circuit voltage: 200V DC depending on pump controller max. voltages
- DC output: 180V DC, 1,500W
- Overload protection: internal fuse, 15A
- Enclosure: steel, gasket-sealed, indoor-use, hinged front cover with key-lock
- Must be protected from direct mid-day sun
- Enclosure class: IP22
- Ambient temperature: max. 45°C
- All wired must be #14 AWG (2.5 mm²) or larger
- Manual with further information is available in partnerNET
- Meets requirements for CE

Electrical Safety



DIMENSION/WEIGHT

- 450 x 300 x 140 mm
17.7 x 11.8 x 5.5 in
- Total weight: 16 kg / 35 lbs



L	N		+	-		+	-	
AC POWER IN (240VAC, 50-60Hz)			SOLAR POWER IN (max. 200VDC)			DC POWER OUT (max. 180VDC)		
GROUND			GROUND			GROUND		

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PP4000 AC PowerPack

AC/DC Converter to Supply PS2-4000 Pump Systems with Backup Power from a Generator or Mains Supply

ORDER INFORMATION

- Item no.: 19-000180 **Product name:** Power Pack 4000 380 V AC
- Item no.: 19-000181 **Product name:** Power Pack 4000 460 V AC

TECHNICAL DATA

Power Pack 4000 380 V AC	Power Pack 4000 460 V AC
3-Phase AC input 380 V ($\pm 20\%$) 45 to 65Hz	3-Phase AC input 460 V ($\pm 15\%$) 45 to 65Hz
DC output: 180-286 V DC 5.7 kW	DC output: 230-330 V DC 6.6 kW
DC output rated current: 20 A at 230 V DC	DC output rated current: 20 A at 330 V DC

- Must be protected from direct mid-day sun
- IP20
- Ambient temperature: -5 to 45°C
- Efficiency up to 94%
- Passive cooling, no fan
- All wires must be #14 AWG (2.5 mm^2) or larger
- Manual with further information is available on PartnerNET
- Meets the requirements for CE

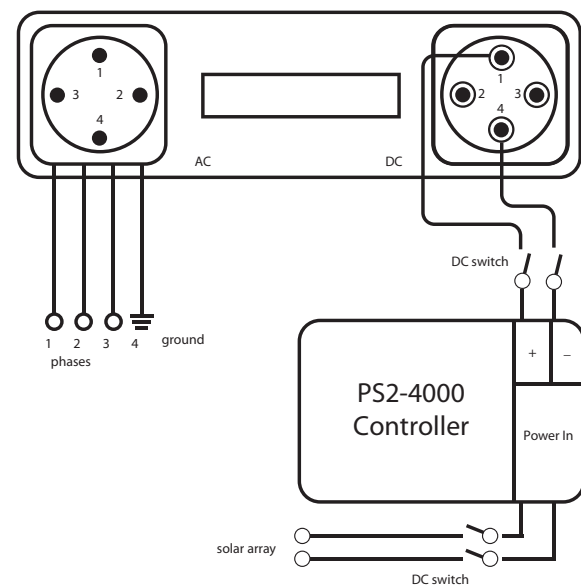
DIMENSION/WEIGHT

- Packaging dimensions: 332 x 210 x 202 mm
15.7 x 8.7 x 8.3 in
- Total weight: 14.5 kg / 30 lbs



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Figure 1: Wiring



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PS BOOST Inline Filter

Inline Filter for use with PS Boost Pumps

PS BOOST Pump Systems are designed for clean water conditions. A PS BOOST Inline Filter must be installed in the suction pipeline to protect the pump from sand, clay or other solids.

ORDER INFORMATION

- **Item no.:** 19-000200 **product name:** Inline Filter 10" PS150 BOOST
- **Item no.:** 19-000210 **product name:** Filter cartridge 10" PS150 BOOST

FEATURES

- Simple to install with included wall holder kit
- Clear bowl for easy maintenance
- Additional spare filter cartridges can be ordered

TECHNICAL DATA

- Inlet: female NPT 1/2"
- Outlet: female NPT 1/2"
- Max. filter cartridge size: 9 7/8 in
- 20-micron spun polypropylene filter cartridges
- Max. water temperature: 50°C (120°F)
- Max. working pressure: 6 bar
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packaging dimensions: 450 x 250 x 250 mm
17.7 x 9.8 x 9.8 in
- Total weight 2.0 kg / 4.4 lbs



PS BOOST Installation Kit

Control unit for use in pressurizing applications

The PS BOOST Installation Kit can be used for any application where a pressure tank is required. If the pressure increases to the maximum, the LORENTZ PS Controller will stop the pump and indicates Tank Full LED.

ORDER INFORMATION

- Item no.: 19-000190 product name: PS BOOST Inst. Kit

FEATURES

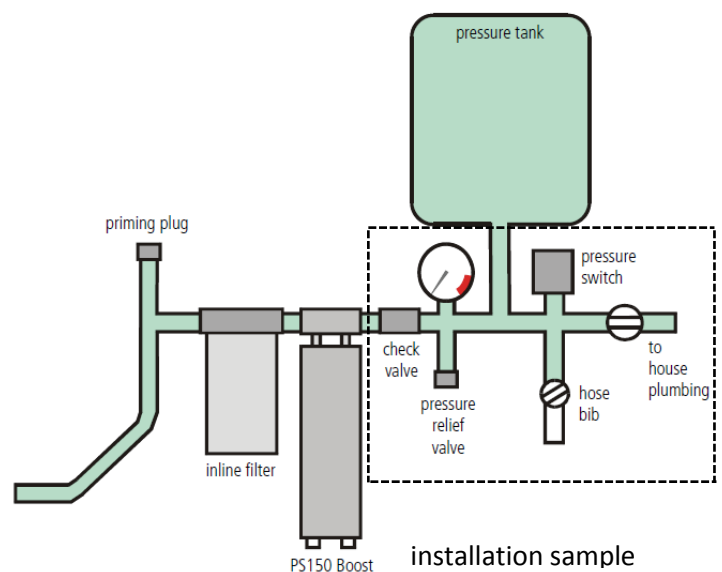
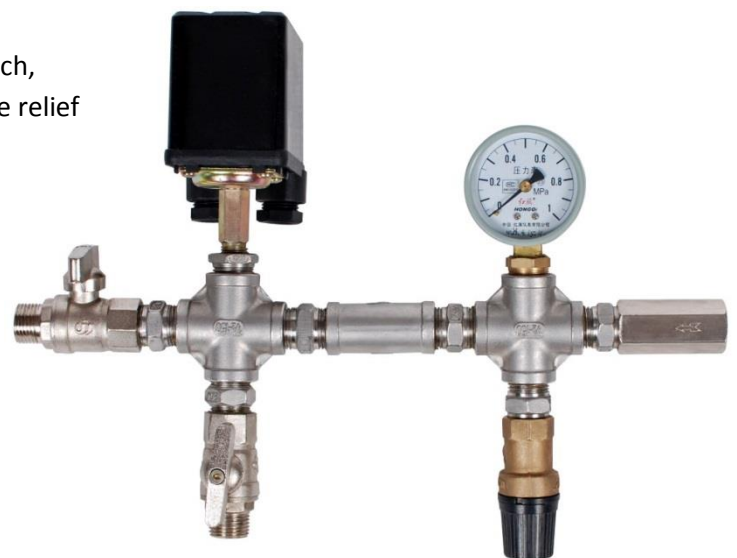
- Unit consist of the LORENTZ Pressure Switch, a check valve, a pressure gauge, a pressure relief valve, two ball valves and two short pipes
- Reliable water pressure detection
- Trouble free operation
- Corrosion-free

TECHNICAL DATA

- Min. cut-in: 1bar
- Max. cut-out: 5bar
- Pressure relief valve: 5.3bar
- Enclosure class: IP44
- Inlet: female G1/2"
- Outlet: male G1/2"
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packing dimensions: 600 x 320 x 170 mm
23.6 x 12.6 x 6.7 in
- Total weight 5.0 kg / 11.0 lbs



PM Accessory bracket

Kit for mounting LORENTZ PS Controller / PV-Disconnect to the Pole Mount PM-1-X

USE / PURPOSE

The PM Accessory bracket is designed for use with the LORENTZ POLE MOUNT-1-X.

The Set allows an easy and flexible mounting of LORENTZ PS Controllers and PV-Disconnect on the Pole of the LORENTZ PM-1-X.



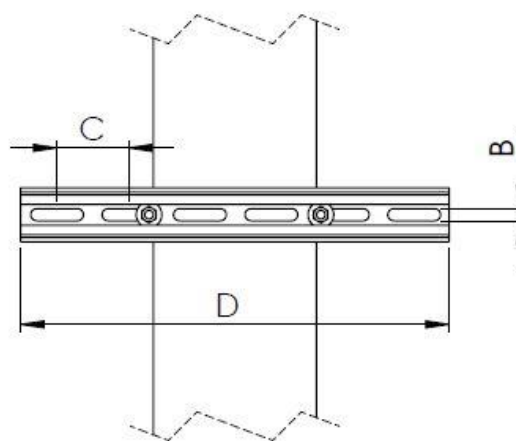
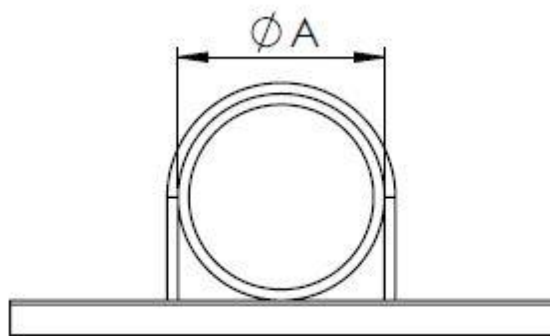
ORDER INFORMATION

Part	Item No.	PM-1-1	PM-1-2	PM-1-3	PM-1-4	PM-1-5	PM-1-6
PM Accessory bracket 2.5-1	19-004220	x	x				
PM Accessory bracket 4-1	19-004210			x	x	x	x

FEATURES

- Simple and cost effective mounting possibility for LORENTZ Controller and PV-Disconnect
- Designed for LORENTZ Pole Mount 1-X

TECHNICAL DATA



	A	B	C	D	Weight	Item no.
PM-1- 1/2	70 -73 mm 2.5 " SCH40	8.5 x 37 mm 0.33 x 1.4 "	50 mm 2 "	300 mm 11.8 "	0.35 kg 0.77 lbs.	19-004220
PM-1- 3/4/5/6	114.3 mm 4 " SCH40				0.4 kg 0.88 lbs.	19-004210

Packing dimensions

305x170x25 mm
12x7x1 inch

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SmartPSUk2-40

AC/DC Converter to Supply PSk2 Pump Systems with Power from a Generator or Grid Supply

Description

The SmartPSUk2-40 automatically blends power from solar and external sources, only using the external source where the demand for water requires it. This is not a switching system that switches between sources but effectively blends power so that only the power that is needed is demanded from the non-solar source. For a hybrid generator solution, this means running at partial load, for a hybrid grid solution this means using kWh from the grid as a top up.

Features

- Provides DC power to PSk2 pump controllers from AC and DC sources
- Blends solar power (DC) with AC power from the grid or generator
- Part of the PSk2 SmartSolution for hybrid powering of water pumping systems
- Data link to PSk2 controller for control and monitoring
- Integrated overheat protection and active cooling



Ordering and shipping information

- Packing dimension: 560 x 400 x 340 mm (22.0 x 15.7 x 13.4 in)
- Packed weight: 18 kg (39 lbs)
- See next page for item numbers and variants

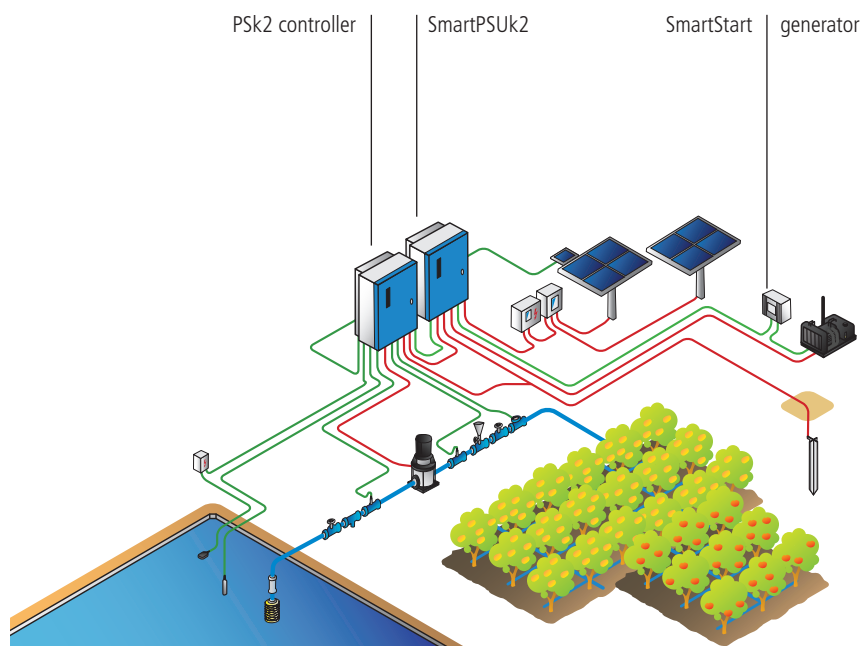
Approvals and standards

The AC/DC converter has the following standards:

- Meets the requirements for CE

photo may differ from actual product

SmartSolution OVERVIEW



BERNT LORENTZ GmbH

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Tel. +49 (0) 4193 8806 - 700, www.lorentz.de

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Order information

Item number	Description
19-002585	SmartPSUk2-40

Mounting options

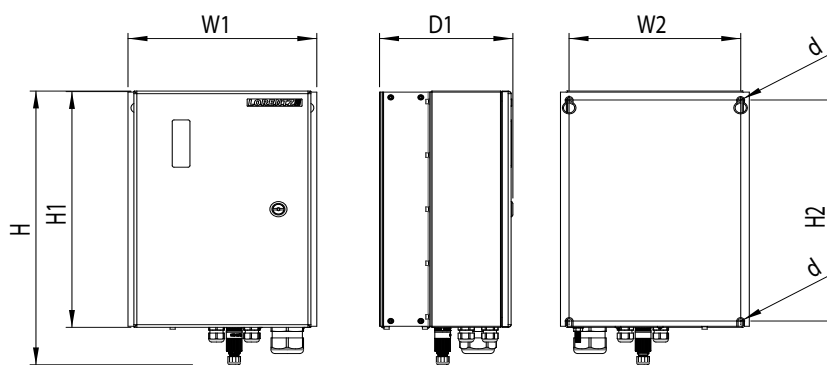
- The SmartPSUk2-40 comes with a cable to connect to the PSk2 Controller (1.00 m) and a back plate for mounting on non-flat surfaces.

Technical data / Specifications

3-Phase AC input	380 – 415 V ($\pm$ 10 %) 50 Hz/60 Hz max. 38 kW (48 kVA)
DC output	U _{max} = 850 V DC I _{max} = 70 A
PV max open circuit voltage:	850 V DC
Environmental protection	IP54
Housing material	Stainless steel powder coated case
Ambient Temperature	-10 to 50°C

Dimensions and weight

- Weight 18kg (39 lbs)
- W1** 320mm (12.6") **W2** 290mm (11.4") **H** 500mm (19.7")
H1 450mm (17.7") **H2** 421mm (16.6") **D1** 226mm (8.9")
d 9mm (0.35")



SCALE 1:10

SCALE 1:10

SCALE 1:10

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SmartStart

Automatic Remote Diesel Generator Switching Device

The SmartStart can automatically and remotely start and stop diesel generators for hybrid solar pumping. It has a secondary ancillary output for switching other devices. It connects to the PSk2 Controller and to the remote switch input of the diesel generator.

ORDER INFORMATION

- Item no.: 19-004280 Product name: SmartStart

FEATURES

- Automatically switches on/off diesel engines equipped with remote start input (NO)
- SmartStart provides power to the PSk2 systems for night time logic operation such as early morning pump starting
- Battery charging from PSk2 Controller
- Shows system status via LEDs
- Ancillary output for switching other devices



TECHNICAL DATA

- Max. contact rating: 277 V AC / 30 V DC / 3 A
- Enclosure class: IP54
- Nominal cross section for GEN Link cable 1.5 mm²
- Further information is available in the PSk2 manual on partnerNET

PRODUCT CONTENT

The SmartStart comes with a cable to connect to the PSk2 Controller. (1.50 m) The required battery must be ordered separately. The battery must meet the following requirements:

- Sealed 12 V AGM lead acid battery Min capacity: 7 Ah

DIMENSION/WEIGHT

- | | |
|-----------------------------------|--|
| • Packaging dimensions: | L x W x H 270 x 250 x 230 mm (10.6 x 9.8 x 9.1 in) |
| • Total weight (without battery): | 2.8 kg (6.2 lbs) |
| • Battery case: | L x W x H 151 x 70 x 95 mm (5.94 x 2.56 x 3.74 in) |

Sun Sensor

Irradiation measurement module and software

The LORENTZ Sun Sensor module and software allow the PS2 and PSk2 controller to accurately estimate irradiation levels and control the pump based on this.

The Sun Sensor function can be configured to offer the pump mechanical protection from repeated starts and stops due to low irradiation. The Sun Sensor function can also be configured to switch external devices via the output terminal.

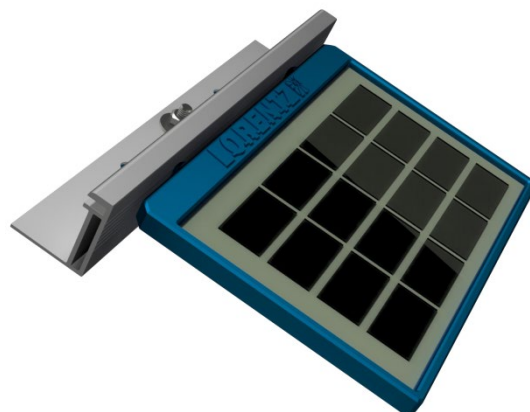
Each module is tested and specially calibrated for use with the PS2 and PSk2 system. The module also includes a software license to use the features which is accessed via the item serial number.

This Sun Sensor module is included with any PS2-1800 or PS2-4000 controller and any PSk2 controller.

For PS controllers (non PS2) refer to the SunSwitch Datasheet.

ORDER INFORMATION

- **Item no.:** 19-005030
- **Product name:** Sun Sensor module



TECHNICAL DATA

- Open circuit voltage $V_{oc} = 10.2 \text{ V}$
- Short circuit current $I_{sc} = 0.19 \text{ A}$
- Wire size: $2 \times 1.0 \text{ mm}^2$ / AWG 18, waterproofed
- Mount in parallel to the solar array
- PS2/PSk2 manual with further information is available in partnerNet
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packaging dimensions: 180 x 205 x 60 mm
7.08 x 8.07 x 2.36 in
- Total weight: 0.6 kg / 1.33 lbs

Flow Sleeve for HR(E) Pumps

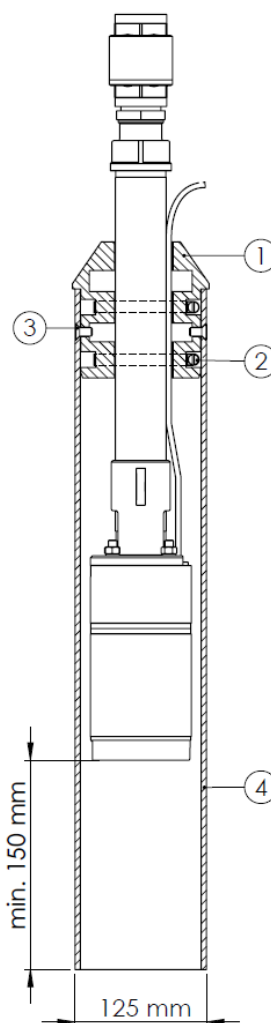
Device to protect the pump from dirt and for motor cooling

A flow sleeve is a sleeve that goes around the pump body and motor. It has two functions:

1. **A flow sleeve reduces the amount of dirt in the pump.** Where water sources contain sand or have the potential for dirt falling into them (uncapped wells) a flow sleeve is recommended to protect the pump from dirt. Due to the large inner diameter of the flow sleeve the water rises very slowly and dirt deposits at the bottom of the well. Subsequently, water with less sediments enters the pump which increases its lifetime.
2. **A flow sleeve cools the motor.** In open water, tanks or pools there is no natural flow of water across the motor which can lead to heat build-up. The tube enforces water flow across the motor which has a cooling effect.

ORDER INFORMATION (please compare (1), (2) etc. with the sample flow sleeve)

Item no.	Product name	Scope of delivery	Suitable for
19-000220	Flow Sleeve Kit for 4" HR Pumps	(1) 1 x flow sleeve adapter (2) 2 x hose clamp (3) 4 x countersunk bolt M6x20 (4) 1 x PVC tube: $\varnothing 125 \times 5$, 800 mm length	All HR pumps with 4" motor
19-000225	Flow Sleeve Kit for 4" HRE Pumps		All HRE pumps with 4" motor
19-000240	Flow Sleeve Adaptor for 4" HR Pumps	(1) 1 x flow sleeve adaptor (2) 2 x hose clamp (3) 4 x countersunk bolt M6x20	All HR pumps with 4" motor
19-000245	Flow Sleeve Adaptor for 4" HRE Pumps	A tube (4) is not part of the scope of delivery.	All HRE pumps with 4" motor



sample flow sleeve

FEATURES

- Due to the large inner diameter of the flow sleeve the water rises very slowly allowing dirt to settle at the bottom of the well
- The life expectancy of the pump is increased due to less sand entering the pump
- Simple installation
- Can be used with PS2-200 – PS2-4000 HR(E) Pumps
- Min. well diameter of 6" (150 mm) required

DIMENSIONS/WEIGHT

Flow Sleeve Kit for 4" HR(E) Pumps (19-000220 or 19-000225):

- Packaging dimensions: 850 x 150 x 130 mm
33.5 x 6.0 x 5.1 in
- Total weight: 3.7 kg / 8.2 lbs

Flow Sleeve Adaptor for 4" HR(E) Pumps (19-000240 or 19-000245, without PVC tube):

- Packaging dimensions: 135 x 135 x 135 mm
5.3 x 5.3 x 5.3 in
- Total weight: 1.2 kg / 2.6 lbs

Flow Sleeve for C-SJ Pumps

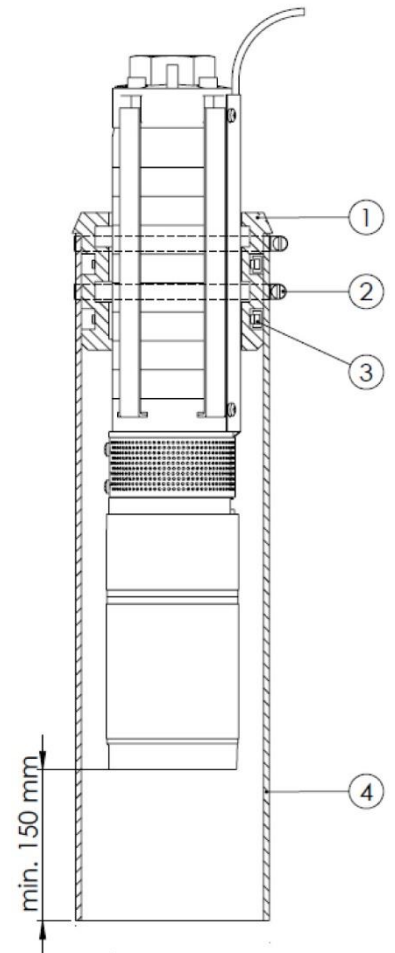
Device to protect the pump from dirt and for motor cooling

A flow sleeve is a sleeve that goes around the pump body and motor. It has two functions:

1. **A flow sleeve reduces the amount of dirt in the pump.** Where water sources contain sand or have the potential for dirt falling into them (uncapped wells) a flow sleeve is recommended to protect the pump from dirt. Due to the large inner diameter of the flow sleeve the water rises very slowly and dirt deposits at the bottom of the well. Subsequently, water with less sediments enters the pump which increases its lifetime.
2. **A flow sleeve cools the motor.** In open water, tanks or pools there is no natural flow of water across the motor which can lead to heat build-up. The tube enforces water flow across the motor which has a cooling effect.

FEATURES

- Due to the large inner diameter of the flow sleeve the water rises very slowly allowing dirt to settle at the bottom of the well
- The life expectancy of the pump is increased due to less sand entering the pump
- Simple installation
- The flow sleeve must be selected according to the pump system (see table)
- The required borehole diameter depends on the diameter of the flow sleeve



ORDER INFORMATION (please compare (1), (2) etc. with the sample flow sleeve)

Item no.	Product name	Scope of delivery	Suitable for
19-000230	Flow Sleeve Kit for 4" C-SJ Pumps with 4" Motor	(1) 1 x adaptor (2) 2 x hose clamp (3) 2 x cable strip (4) 1 x PVC tube: D140x4, 0.5 m length	All 4" C-SJ pump ends with 4" motor
19-000250	Flow Sleeve Adaptor for 4" C-SJ Pumps with 4" Motor	(1) 1 x adaptor (2) 2 x hose clamp (3) 2 x cable strip A PVC tube D140x4mm (4) is not part of the scope of delivery.	All 4" C-SJ pump ends with 4" motor
19-000255	Flow Sleeve Adaptor for C-SJ17-X/30-X with 4" motor	(1) 1 x adaptor (2) 2 x screw 4.2x16 (3) 1 x hose clamp A PVC tube D160x4mm (4) is not part of the scope of delivery.	All C-SJ17-X/30-X with 4" motor
19-005170	Flow Sleeve Kit for C-SJ42-X Pumps	(1) 1 x adaptor (2) 2 x hose clamp d =190-210 mm (3) 2 x hose clamp d= 150-170 mm (4) 1 x PVC tube: D200x5, 1.4 m length	All C-SJ42-X pumps
19-005180	Flow Sleeve Adaptor for C-SJ42-X Pumps	(1) 1 x adaptor (2) 2 x hose clamp d=190-210 mm (3) 2 x hose clamp d=150-170 mm A PVC tube D200x5mm (4) is not part of the scope of delivery.	All C-SJ42-X pumps
19-001400	Flow Sleeve Kit for C-SJ8-x/12-x for 6" motor with tube	(1) 1x adaptor (2) 2x hose clamp d=190-210 mm (3) 2x hose clamp d=150-170 mm (4) 1x PVC tube: D200x5, 1,4m length	All C-SJ8-x and C-SJ12-x pumps with 6" motor
19-001450	Flow Sleeve Adaptor for C-SJ8-x/12-x for 6" motor	(1) 1x adaptor (2) 2x hose clamp d=190-210 mm (3) 2x hose clamp d=150-170 mm A PVC tube D200x5mm (4) is not part of the scope of delivery.	All C-SJ8-x and C-SJ12-x pumps with 6" motor
19-001410	Flow Sleeve Kit for C-SJ17-x/30-x for 6" motor with tube	(1) 1x adaptor (2) 2x hose clamp d=190-210 mm (3) 2x hose clamp d=150-170 mm (4) 1x PVC tube: D200x5, 1,4m length	All C-SJ17-x and C-SJ30-x pumps with 6" motor
19-001460	Flow Sleeve Adaptor for C-SJ17-x/30-x for 6" motor	1x adaptor 2x hose clamp d=190-210 mm 2x hose clamp d=150-170 mm A PVC tube D200x5mm (4) is not part of the scope of delivery.	All C-SJ17-x and C-SJ30-x pumps with 6" motor

DIMENSIONS/WEIGHT

Flow Sleeve Kit for 4" C-SJ Pumps with 4" Motor (19-000230):

- Packaging dimensions: 590 x 150 x 130 mm
23.2 x 6.0 x 5.1 in sample flow sleeve
- Total weight: 3.3 kg / 7.3 lbs

Flow Sleeve Adaptor for 4" C-SJ Pumps with 4" Motor (19-000250, without tube):

- Packaging dimensions: 165 x 165 x 110 mm
6.5 x 6.5 x 4.3 in
- Total weight: 1.2 kg / 2.6 lbs

Flow Sleeve Adaptor for C-SJ17-X/30-X with 4" motor (19-000255, without tube):

- Packaging dimensions: 165 x 165 x 110 mm
6.5 x 6.5 x 4.3 in
- Total weight: 1.2 kg / 2.6 lbs

Flow Sleeve Kit for C-SJ42-X Pumps (19-005170):

- Packaging dimensions: 1500 x 205 x 215 mm
59.0 x 8.1 x 8.5 in
- Total weight: 9.3 kg / 20.6 lbs

Flow Sleeve Adaptor for C-SJ42-X Pumps (19-005180, without tube):

- Packaging dimensions: 230 x 230 x 130 mm
9.1 x 9.1 x 5.1 in
- Total weight: 1.9 kg / 4.2 lbs

Flow Sleeve Kit for C-SJ8-X/12-X Pumps (19-001400):

- Packaging dimensions: 1510 x 215 x 235 mm
59.4 x 8.4 x 9.2 in
- Total weight: 11 kg / 24.3 lbs

Flow Sleeve Adaptor for C-SJ8-X/12-X Pumps (19-001450, without tube):

- Packaging dimensions: 220 x 220 x 130 mm
8.6 x 8.6 x 5.1 in
- Total weight: 3 kg / 6.61 lbs

Flow Sleeve Kit for C-SJ17-X/30-X Pumps (19-001410):

- Packaging dimensions: 1510 x 215 x 235 mm
59.4 x 8.4 x 9.2 in
- Total weight: 11 kg / 24.3 lbs

Flow Sleeve Adaptor for C-SJ17-X/30-X Pumps (19-001460, without tube):

- Packaging dimensions: 220 x 220 x 130 mm
8.6 x 8.6 x 5.1 in
- Total weight: 3 kg / 6.61 lbs

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PS2 Manual Speed Controller

Device to provide manual motor speed control of PS2 systems

The LORENTZ PS2 Manual Speed Controller allows adjustment of the maximum motor speed without using the PumpScanner App. To use the Manual Speed Controller, it is required to activate this function in the settings of PumpScanner during or before installation.



ORDER INFORMATION

- Item no.: 19-000035 Product name: PS2 Manual Speed Controller

FEATURES

- Allows manual control of PS2 motor speed
- Outdoor rated, installed in the housing of the controller

TECHNICAL DATA

- Voltage: 15-24 V DC
- Enclosure class: IP65
- Ambient temperature: -38...50 °C (-36... 122 F)
- Wire size: 2 x 0,75 mm²/18 AWG
- Replaces Ø 20mm cable gland
- Meets the requirements for CE
- Please note that if "Manual Speed Controller" is configured then "Set speed limitation" function is not available in PumpScanner.

DIMENSION/WEIGHT

- Packing dimension : 100 x 70 x 35 mm; 3.9 x 2.7 x 1.3 in
- Total weight: 0.2 kg / 0.4 lbs



PV Disconnect 1000-300-30

30 string connection box with DC disconnect switch and surge protection

ORDER INFORMATION

- Item no.: 19-000113
- Product name: PV Disconnect 1000-300-30

FEATURES

- Simple and cost-effective unit to connect up to 30 module strings in parallel
- Includes an appropriate DC rated disconnect switch and string fuses (15 A)
- Includes surge protection
- Designed for PSk2-100 pump systems

TECHNICAL DATA

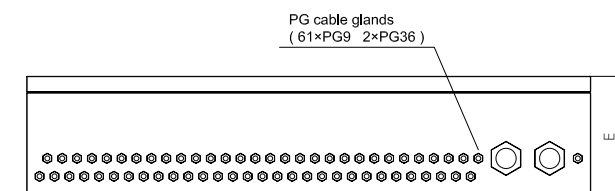
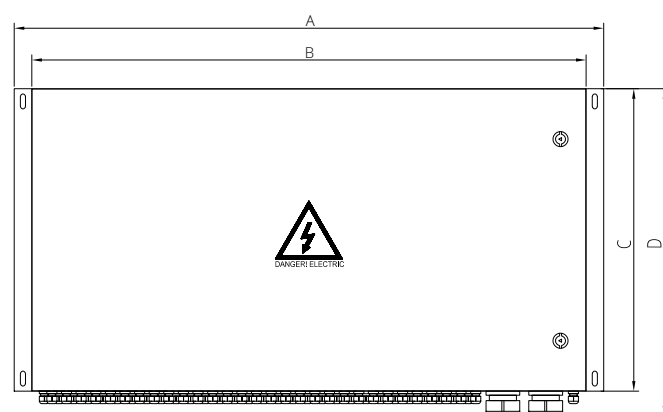
- Wiring of up to 30 PV strings in parallel
- DC rated disconnect switch enclosed
- Enclosure class IP 54
- Meets the requirements for CE

Max. voltage	1,000 V DC (U_{oc}) 880 V DC (U_{mp})
Max. current per string	15 A
Max. total current	300 A
Max. no. of strings	30
String cable size	4 - 10 mm ²
Output cable size	4 - 16 mm ²



photo may differ from actual product

DIMENSION/WEIGHT [mm]



Dimension in mm [in]					Weight in kg [lbs]
A	B	C	D	E	
1170	1100	600	650	230	68
[46.06]	[43.31]	[23.62]	[25.60]	[9.06]	[149.91]

Packing dimensions:

1240 mm x 800 mm x 410 mm (48.82 in x 31.50 in x 16.14 in)

Weight

90 kg (198.42 lbs)

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PS 2 Controller Plug Kit

Kit for an easy and electrical safe installation of PS2 Controller

The LORENTZ PS2 Controller Plug Kit can be installed on any PS2 system. The kit extends the internal wiring connections to plugs allowing systems to be pre-wired and delivered to site. Possible uses for the plug kit are where time on site needs to be minimized, where systems are often moved or where the skills available onsite do not allow for a standard installation.

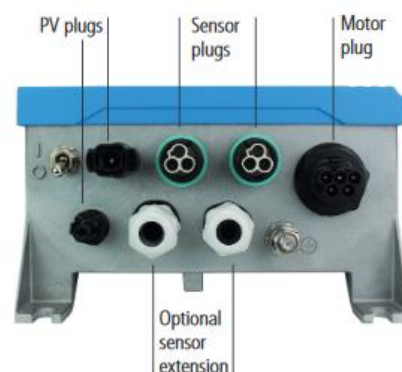
The sensor extension kit allows the installation of two additional sensors, the standard set contains only two plug sets for sensor connections.

Order Information

- 19-005001 Plug- Kit PS2- Controller
- 19-005011 Sensor Plug Extension Kit

Features

- Allows fast, easy and electrical safe installation of PS2 Controller
- Customer must not open the controller for installation
- Outdoor rated, all parts are designed for outdoor use



Technical Data

PLUG	Wire size	max. current	max. voltage	Ambient temperature
Motor	max, 6mm ² (10 AWG)	32 A	600V DC	-40°C ... +90°C
PV	max 8mm ² (8 AWG)	40 A (27A at 2.5mm ² /14AWG)	1500V DC	-40°C ...+90°C
Sensor	max. 1.5mm ² (16 AWG)	3 A	50V DC	-40°C ...+90°C

Packing Dimension/Weight

19-005001 Plug- Kit PS2- Controller

- Packing dimension: 16 x 300 x 4cm (6.3x12x1.6inch)
- Total weight: 0.33kg (0,73lb)

19-005011 Sensor Plug Extension Kit

- Packing dimension: 16 x 24 x 2cm (6.3x10x0.8inch)
- Total weight: 0.1kg (0,22lb)



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PS2-100 Well Probe

Mechanical float switch for run-dry protection of PS2-100 solar pumps

The PS2-100 Well Probe provides a reliable method of run-dry protection. It detects that water is present within a well, tank or other water source. The well probe is typically attached to the riser pipe above the pump and connected to the controller. When the well probe becomes dry (water level is below the probe) the pump switches off to avoid dry running.

ORDER INFORMATION

Item no.: 19-000007 **Product name:** Well Probe with 15 m / 49 ft cable and a XT-30F plug

FEATURES

- Reliable run-dry protection
- Simple to install using pre-fitted plug

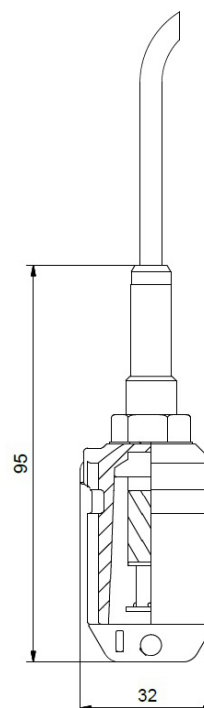


TECHNICAL DATA

- Max. operating temperature 55°C / 131°F
- Enclosure class: IP68
- Submersion depth: max 50 m / 164 ft
- Wire size: 2 x 0.75 mm² or AWG 19, waterproofed
- 15 m / 49 ft cable with XT-30F plug
- Must be mounted in a vertical position
- Meets the requirements for CE

DIMENSION / WEIGHT

- Packaging dimensions: 205 x 160 x 80 mm
8.1 x 6.3 x 3.1 in
- Total weight: 0.85 kg / 1.9 lbs



PS2-100 Motor Extension Cable

Cable with plugs to extend the motor cable on PS2-100 systems

This 15 m / 50 ft cable is used to extend the motor cable on PS2-100 pump systems. The cable has a 3 pin MT-60M connector on one end and a MT-60F connector on the other end. The extension cable comes with a waterproof connector tube which seals the cable connection and provides strain relief.

ORDER INFORMATION

- **Item no.:** 19-001220
- **Product name:** PS2-100 15 m / 50 ft Motor Extension Cable

FEATURES

- Cable with plugs
- Simple to install
- Includes waterproof connector tube



TECHNICAL DATA

- Operating temperature: -10°C to 55°C / 14°F to 131°F
- Storage temperature: -10°C to 55 °C / 14°F to 13°F
- Cable length: 15 m / 50 ft, waterproof
- Wire size: 3 x 2.5 mm² or AWG 14
- Meets the requirements for CE

DIMENSION / WEIGHT

- Packaging dimensions: 385 x 385 x 130 mm
15.1 x 15.1 x 5.1 in
- Total weight: 3.6 kg / 7.9 lbs

PS2-100 Float Switch

Mechanically activated device for water level detection

The PS2-100 Float Switch can be used to detect the water level in a tank. When the water level in the tank reaches the maximum the LORENTZ PS2-100 Controller will stop the pump and indicates tank full LED.

ORDER INFORMATION

- **Item no.:** 19-000032 **product name:** PS2-100 Float Switch with 10 m / 33 ft cable

FEATURES

- NC (normally closed) function
- Reliable water level detection
- Simple plug-in installation
- Trouble-free operation
- Not sensitive to rotation
- Corrosion-free

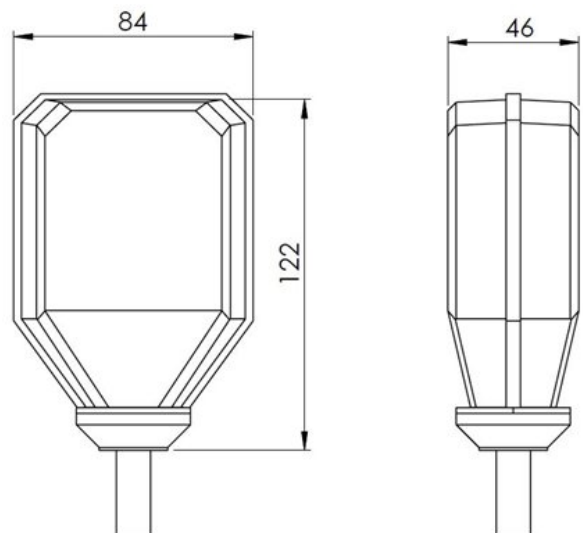


TECHNICAL DATA

- Operating temperature: -10°C to 55°C / 14°F to 131°F
- Storage temperature: -10°C to 55°C / 14°F to 131°F
- Enclosure class: IP68
- Cable length: 10 m / 33 ft, waterproof
- Meets the requirements for CE

DIMENSION / WEIGHT

- Packaging dimensions: 250 x 205 x 105 mm
9.8 x 8.1 x 4.1 in
- Total weight: 1.9 kg / 4.2 lbs



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Surge Protector2

Device to Protect LORENTZ Pump Accessories from Voltage Spikes

ORDER INFORMATION

- Item no.: 19-005210 product name: Surge Protector2

FEATURES/COMPATIBILITY

- Reliable surge protection device for any switched, pulse or analogue (4-20 mA) inputs sensors including:
 - Well Probe Sensor 19-000000
 - Water Sensor 19-000001
 - Float Switch 19-000030
 - Pressure Switch 19-000310
 - Liquid Level (all types, e.g. 19-005040)
 - Liquid Pressure Sensor (all types, e.g. 19-004460)
 - Water Meter (all types, e.g. 19-002160)
 - Sun Switch (19-000050)
- The device must be installed inside the PS2 or PSk2 controller.



TECHNICAL DATA

- Max. voltage: 30 V DC
- Max current 8/20μs: 500 A
- Enclosure class: IP20
- Ambient temperature: max. 80°C (176°F)
- Wire size: 2 x 1.5mm² (AWG 16)
- Meets the requirements for CE

DIMENSION/WEIGHT

- Packing dimensions: 56 x 26 x 120 mm
 2.2 x 1.02 x 0.47 in
- Total weight 0.1 kg / 0.2 lbs

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PS2-100 Extension Cable for Well Probe / Float Switch

Cable with plugs to extend the well probe / float switch cable on PS2-100 systems

This 15 m / 50 ft cable is used to extend the well probe cable on the PS2-100 pump system. The cable has a 2 pin XT-30M connector on one end and a XT-30F connector on the other end. The extension cable comes with a waterproof connector tube which seals the cable connection and provides strain relief.

ORDER INFORMATION

- **Item no.:** 19-001240
- **Product name:** PS2-100 15 m / 50 ft Well Probe / Float Switch Extension Cable

FEATURES

- Cable with plugs
- Simple to install
- Includes waterproof connector tube

TECHNICAL DATA

- Operating temperature: -10°C to 55°C / 14°F to 131°F
- Storage temperature: -10°C to 55°C / 14°F to 131°F
- Enclosure class: IP68
- Cable length: 15 m / 50 ft, waterproof
- Wire size: 2 x 1 mm² or AWG 17
- Meets the requirements for CE

DIMENSION / WEIGHT

- Packaging dimensions: 250 x 205 x 105 mm
8.1 x 8.1 x 4.1 in
- Total weight: 0.7 kg / 1.5 lbs



Retrofit Adaptors for HR/HRE Pump

Adaptor set to combine HR and HRE pump design

PURPOSE

LORENTZ introduced the next evolution of the helical rotor pump, called HRE. This HRE design has different connection dimension than the previous HR design. In case an ECDRIVE or a Pump End of the HRE generation shall be combined with a component of the previous HR generation a retrofit adaptor set must be used.

ORDER INFORMATION

Item no.	Product name	Content of delivery	Purpose
19-001820	HRE Retrofit Set A		to connect a ECDRIVE HRE with a PE HR
19-001830	HRE Retrofit Set B		to connect an PE HRE with an ECDRIVE HR

Packing Dimensions/Weight

19-001820 HRE Retrofit Set A

- Packing dimensions: 60 x 20 x 20 mm / 2.4 x 0.8 x 0.8 in
- Total weight 0.2 kg / 0.4 lbs

19-001830 HRE Retrofit Set B

- Packing dimensions: 430 x 26 x 26 / 16.9 x 1 x 1 in
- Total weight 0.3 kg / 0.7

SmartPSUk2-100

AC/DC Converter to Supply PSk2 Pump Systems with Power from a Generator or Grid Supply

Description

The SmartPSUk2-100 automatically blends power from solar and external sources, only using the external source where the demand for water requires it. This is not a switching system that switches between sources but effectively blends power so that only the power that is needed is demanded from the non-solar source. For a hybrid generator solution, this means running at partial load, for a hybrid grid solution this means using kWh from the grid as a top up.

Features

- Provides DC power to PSk2 pump controllers from AC and DC sources
- Blends solar power (DC) with AC power from the grid or generator
- Part of the PSk2 SmartSolution for hybrid powering of water pumping systems
- Data link to PSk2 controller for control and monitoring
- Integrated overheat protection and active cooling



Ordering and shipping information

- Packing dimension: 580 x 740 x 990 mm (22.8 x 29.1 x 40.0 in)
- Packed weight: 65 kg (144 lbs)
- See next page for item numbers and variants

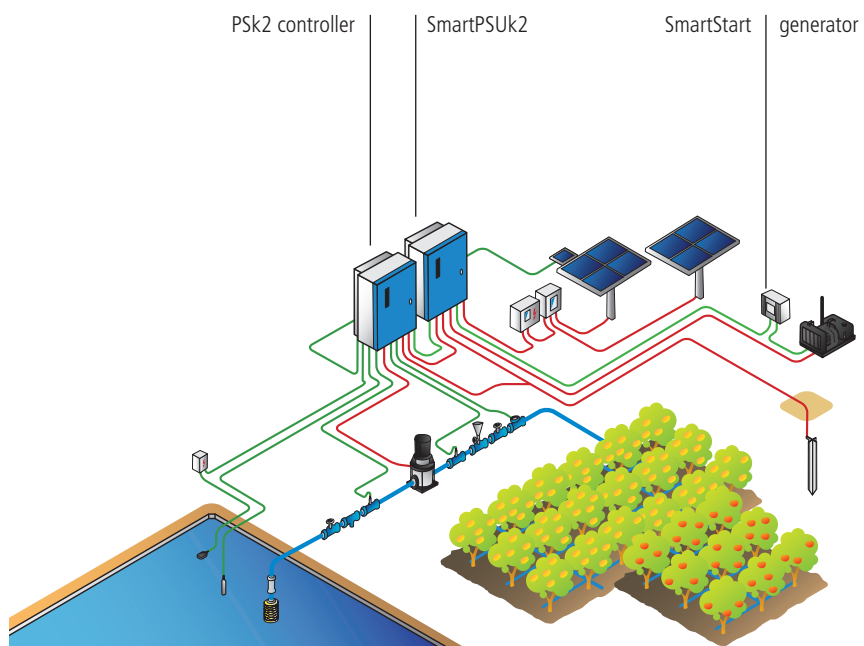
Approvals and standards

The AC/DC converter has the following standards:

- Meets the requirements for CE

photo may differ from actual

SmartSolution OVERVIEW



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All specifications and information are given with good intent, errors are possible and products may be subject to change without notice. Pictures may differ from actual products depending on local market requirements and regulations.

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Order information

Item number	Description
19-002570	SmartPSUk2-100

Mounting options

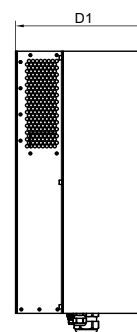
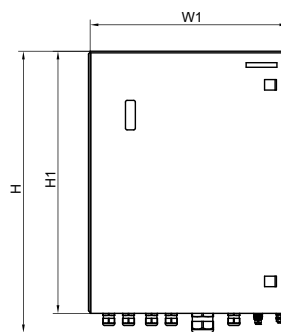
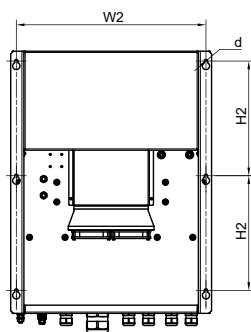
- The SmartPSUk2-100 comes with a cable to connect to the PSk2 Controller (3.50 m) and a back plate for mounting on non-flat surfaces.

Technical data / Specifications

3-Phase AC input	380/400/415 VAC 50 Hz/60 Hz max. 95 kW
DC output	U _{max} = 850 V DC I _{max} = 170 A
PV max open circuit voltage:	850 V DC
Environmental protection	IP54
Housing material	Stainless steel powder coated case
Ambient temperature	-10 to 50°C

Dimensions and weight

- Weight 65 kg (143.3 lbs)
- W1** 610mm (24") **W2** 576mm (22.6") **H** 859mm (33.8")
H1 800mm (31.5") **H2** 350mm (13.8") **D1** 402mm (15.8")
d 10mm (0.4")



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PowerPack 150S

DC power supply for PS2 pump systems

Description

An outdoor rated power supply unit to allow LORENTZ PS2-100 to PS2-150 systems to be powered from an AC (mains voltage) power source. The PowerPack can be used as a continuous supply, in conjunction with PV modules or as a temporary or seasonal supply.

Features

- DC power supply to power LORENTZ solar pump systems from the grid or a generator
- Robust weatherproof housing
- External on / off switch
- Use of MC4 connections to allow quick change over from PV modules to PowerPack
- DC inputs and outputs allow for permanent connection of a solar array
- 1m AC connection cable - NEMA 5-15P (US Plug) or CEE 7/7 (EU Plug)
- Requires a tool to open the housing
- Designed to be used with the LORENTZ PS2-100 to PS2-150 systems



photo may differ from actual product

Ordering and shipping information

- Packed volume: 0.03 m³ (1.03 ft³)
- Packed weight: 3.6 kg (7.1 lbs)
- See next page for item numbers and variants

Approvals and standards

The switch mode power supply has the following approvals

- EN62368-1
- IEC 62103
- IEC 61000-6-2
- IEC 61000-6-4
- AS/NZS 60950-1
- UL62368-1



Technical data / Specifications

	PowerPack 150S
For use with these Pump systems	PS2-100 PS2-150
AC input	100-260 V AC / 50 - 60 Hz
DC output	24 V - 13.4 A
Maximum PV input	50 V DC
PV in connections	2 x MC4
DC out connections	2 x MC4
AC input	IEC C14 female (US or EU cable supplied)
Environmental protection	IP23
Housing material	Polycarbonate

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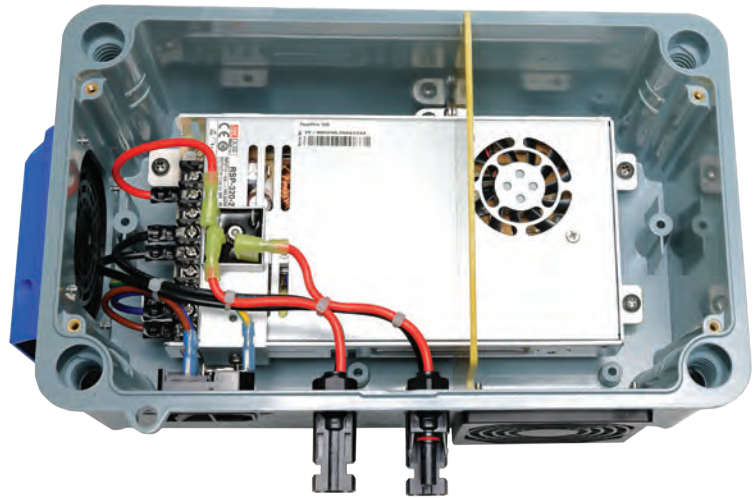
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Order information

Item number	Description
19-005290	PowerPack 150S, EU
19-005295	PowerPack 150S, US

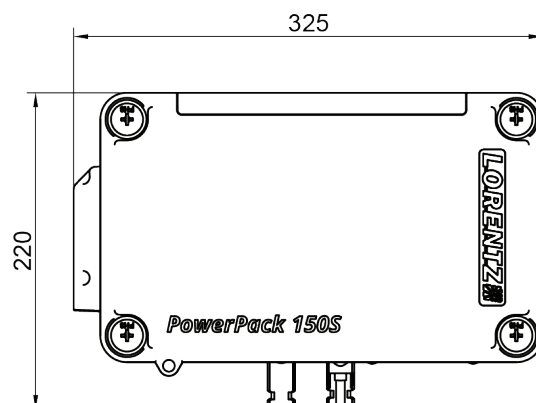
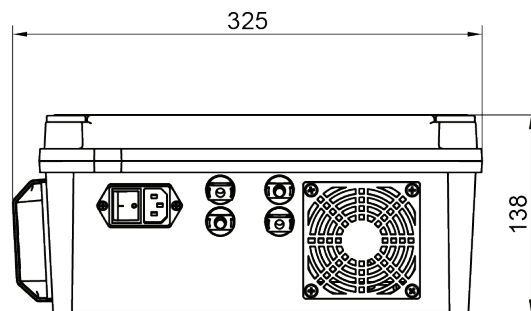
Mounting options

- Wall or panel mount using 4 holes with weather protection



Dimensions and weight

- See drawing for mm measurements
- Max height 220 mm (8.7")
- Max width 325 mm (12.8")
- depth 138 mm (5.4")
- Weight 2.9 kg (6.4 lbs)



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